Signal Recovery from Raw Options Chains: Evidence that Standard GEX Metrics Discard Structural Information

Christopher Regan* Ying Xie†

Abstract

Standard gamma exposure metrics compress rich options chain data into scalar values, potentially discarding valuable structural signal. We demonstrate that large language models processing raw strike-level data achieve 92.3% detection of dealer hedging constraints—outperforming the same models given pre-calculated Net GEX by 30.8 percentage points. This finding establishes that unstructured feature extraction recovers signal lost in parametric compression. Using obfuscation testing—stripping all temporal context while preserving structural relationships—we validate genuine pattern detection rather than training data memorization. Across 242 trading days, 91.2% of detected patterns materialize in subsequent price movements. Critically, detection capability remains stable (68–74%) while trading profitability collapses to zero, establishing these signals as risk management indicators rather than alpha generators. Detection days exhibit 33% lower range expansion than baseline, proving the model identifies volatility suppression mechanisms. Our methodology provides rigorous validation standards for deploying AI-based structural analysis in financial risk systems.

Highlights:

- Raw options chain data yields 30.8 percentage points higher detection than pre-calculated Net GEX, demonstrating that standard parametric features discard structural signal recoverable through unstructured processing.
- Obfuscation testing—removing all temporal context—validates genuine structural pattern detection versus training data memorization, achieving 91.2% materialization accuracy across 242 trading days.
- Stable detection (68–74%) despite zero trading alpha establishes dealer hedging signals as risk management indicators rather than alpha generators, with detected days showing 33% lower volatility.

Keywords: gamma exposure, dealer hedging, market microstructure, feature extraction, signal compression, obfuscation testing, risk management, large language models

JEL Classification: G12 (Asset Pricing), G14 (Information and Market Efficiency), C45 (Neural Networks), C55 (Large Data Sets)

*Department of Computer Science, Kennesaw State University, Marietta, GA. Email: cregan1@kennesaw.edu

†Department of Information Technology, Kennesaw State University, Marietta, GA. Email: yxie2@kennesaw.edu

1 Introduction

Options market makers operate under regulatory mandates to maintain delta-neutral books despite accumulating gamma exposure (U.S. Securities and Exchange Commission 2024). When aggregate gamma exposure turns substantially negative, dealers face forced pro-cyclical hedging—selling strength and purchasing weakness—thereby amplifying price movements through coordinated rebalancing flows (Ni et al. 2005; Gârleanu et al. 2009). Practitioners commonly summarize these dynamics through net gamma exposure (GEX) metrics, compressing rich strike-level options data into scalar values for tractability. Yet a critical question remains: does this parametric compression discard structural signal relevant for detecting dealer constraints?

We demonstrate that it does. Large language models processing raw strike-level data—open interest, implied volatility, and bid-ask spreads per strike—achieve 92.3% detection of dealer hedging patterns, outperforming identical models given pre-calculated Net GEX by 30.8 percentage points. This finding establishes that standard industry metrics represent lossy compression of actionable structural information. The options surface contains signal about dealer positioning that scalar GEX values cannot capture: asymmetrical strike concentrations, volatility skew patterns, and spatial distribution of open interest relative to spot price.

This result emerges from a rigorous validation methodology we term *obfuscation testing*. Presenting a model with "SPY on January 27, 2021" alongside substantial negative gamma risks measuring memorization of the GameStop squeeze rather than comprehension of underlying dealer mechanics. This training data contamination problem pervades machine learning evaluation in finance, where temporal context enables recall rather than reasoning. Obfuscation testing eliminates all memorizable information while preserving structural relationships: converting dates to generic labels ("Day T+0"), anonymizing ticker symbols ("SPY" $\rightarrow$ "INDEX_1"), and removing economic event references. Successful pattern detection under these conditions validates genuine structural analysis rather than statistical correlation with training data.

The proliferation of zero-days-to-expiration (0DTE) options establishes a particularly rigorous validation environment. Trading volume in 0DTE contracts expanded from negligible levels pre-2022 to exceeding 50% of total SPX options volume by 2024 (CBOE Global Markets 2023), generating a persistent negative gamma regime where dealers maintain net short gamma across 95.6% of trading days in our sample. This structural transition from historical alternating regimes (Ni et al. 2005; Gârleanu et al. 2009) to sustained single-regime dynamics presents a demanding test case: unlike conventional pattern recognition exploiting regime variation, our methodology must identify the underlying constraint mechanism within a uniform environment.

Our methodology advances financial data science through four contributions. First, we establish that unstructured feature extraction from raw options chains recovers signal lost in parametric GEX compression—a 30.8 percentage point detection improvement with direct implications for risk system design. Second, we introduce obfuscation testing as a validation framework for distinguishing genuine structural pattern detection from training data memorization. Third, we quantify prompt sensitivity effects: including regime labels ("NEGATIVE_GAMMA") inflates detection from 71.5% to 100%, revealing how contextual hints generate misleading performance metrics. Fourth, we establish that detected patterns function as risk management signals rather than alpha generators: stable detection capability (68-74% quarterly) persists as economic profitability collapses (Sharpe 1.8 $\rightarrow$ 0.1), and detection days exhibit 33% lower volatility than baseline—proving identification of suppression mechanisms rather than high-volatility events.

Testing this framework on 242 trading days throughout 2024, we evaluate three manifestations of dealer hedging constraints: gamma positioning, stock pinning, and 0DTE hedging. Using fully unbiased prompts with obfuscated data, we achieve 71.5% average detection rate, with 91.2% of detected patterns materializing in forward returns. The WHO→WHOM→WHAT causal framework requires articulation of economic actors, affected parties, and forced mechanisms, ensuring detected patterns reflect genuine market microstructure dynamics.

These findings have implications for both risk management and AI deployment in finance. For practitioners, our results demonstrate that standard GEX metrics underutilize available structural information, suggesting opportunities to improve risk system inputs through unstructured feature extraction. For researchers, obfuscation testing offers a rigorous framework for validating pattern detection in domains where training data contamination is a concern.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents our methodology. Section IV describes the experimental setup. Section V reports results. Section VI discusses implications and limitations. Section VII concludes.

2 Background and Related Work

2.1 Dealer Hedging and Market Microstructure

Market makers face regulatory requirements to maintain delta-neutral positions (U.S. Securities and Exchange Commission 2024; Financial Industry Regulatory Authority 2024), creating systematic hedging flows when gamma exposure accumulates. The theoretical foundation for these mechanics was established by Grossman and Miller (1988), who demonstrated how market maker inventory risk creates predictable hedging patterns. Frey and Stremme (1997) formalized how dynamic hedging amplifies volatility through feedback effects, while Avellaneda and Lee (2010) showed these patterns create statistical arbitrage opportunities.

A critical market microstructure principle underlying our methodology is the dealer counterparty framework: market makers serve as counterparties to customer option positions, such that dealer gamma equals the negative of aggregate customer gamma. Anderegg et al. (2022) establishes this relationship theoretically, showing that all option trades must be attributable to either market takers or market makers, with positions summing to zero. Empirically, Dim et al. (2025) validates this framework by measuring market maker inventory directly from order flow, demonstrating that market maker gamma is systematically opposite to customer demand.

Empirically, Ni et al. (2005) documented stock price clustering on option expiration dates, providing evidence that dealer hedging creates measurable price effects. Gârleanu et al. (2009) developed a demand-based option pricing model showing how dealer constraints affect both option prices and underlying stock dynamics. More recently, Ge et al. (2016) demonstrated that the option-to-stock volume ratio predicts future returns, validating the economic significance of options market mechanics.

2.2 Gamma Exposure in Practice

Practitioner research has operationalized gamma exposure metrics through standardized calculations (SpotGamma Research 2019, 2021; SqueezeMetrics 2017, 2020), with major investment banks now publishing regular research on gamma hedging flows (McElligott 2017). The rise of zero-days-to-expiration (0DTE) options has intensified these effects (CBOE Global Markets 2023; JPMorgan Markets Research 2023), making dealer hedging pattern detection increasingly important for understanding price dynamics.

2.3 GEX Limitations and Robustness

Practitioner gamma exposure (GEX) calculations rely on a simplifying assumption: that aggregate customer positions are net short calls and net long puts. This assumption generally holds for broad index options where institutional hedging demand dominates; however, it may diverge during periods of speculative buying or distinctive customer positioning shifts. Krishnan and Bennington (2021) document how dealer delta hedging, while the primary defense against gamma risk, creates feedback loops and cascading volatility effects that can destabilize markets. Traders have noted that Options Depth (actual dealer positioning from order flow) can differ materially from GEX estimates, particularly during market stress or extreme volatility.

Our analysis acknowledges this limitation explicitly: we employ GEX as a practical proxy for dealer gamma exposure due to data availability constraints, not as definitional truth. Critically, our LLM detection framework is robust to moderate GEX variations because validation occurs through forward-return materialization in unbiased obfuscation testing (91.2% prediction materialization rate with fully obfuscated temporal and ticker data), not GEX accuracy. If actual dealer positioning diverges from our GEX estimates, such divergences represent measurement artifacts, not invalidations of the underlying mechanism. Dealers must hedge their options exposure to maintain delta neutrality regardless of whether gamma is measured from order flow, open interest, or alternative methodologies. Thus, detected patterns reflect fundamental market mechanics rather than artifacts of any particular gamma estimation formula.

2.4 Rationale for Gamma-Centric Analysis

We deliberately constrain our analysis to gamma exposure rather than incorporating higher-order Greeks (vanna, charm, vomma) for both theoretical and practical reasons that strengthen rather than limit our validation framework.

Industry Fragmentation of Higher-Order Models. While academic literature provides standardized formulas for vanna ($\partial^2 V/\partial S \partial \sigma$) and charm ($\partial^2 V/\partial S \partial t$) (Ederington and Guan 2007), institutional implementations vary significantly. Major dealers employ proprietary adjustments for volatility surface dynamics, jump risk, and microstructure effects (Mixon 2009). This heterogeneity makes "ground truth" validation impossible—we cannot verify LLM detection against an objective standard when no such standard exists. Ederington and Guan (2007) demonstrate that while higher-order Greeks improve price prediction, their practical implementation remains institution-specific.

Gamma as Universal Constraint. Unlike higher-order Greeks, gamma hedging represents a regulatory requirement universally binding across institutions (Mixon 2009). Dealers must maintain delta-neutral books regardless of their internal vanna models. This universality makes gamma exposure the ideal testing ground: all market participants face the same fundamental constraint, even if their precise calculations differ marginally. The primacy of gamma in dealer risk management is well-established in practitioner literature (Pearson 2013).

Observability and Materialization. Gamma effects manifest clearly in price action—negative gamma creates observable volatility amplification measurable through forward returns (Ni et al. 2022). Vanna and charm effects, conversely, entangle with numerous confounding factors (correlation changes, term structure shifts) that obscure validation. By focusing on gamma, we can definitively verify whether detected patterns materialize.

Avoiding Overfitting to Proprietary Models. Training or testing LLMs on higher-order Greeks risks overfitting to specific institutional implementations. A model detecting "vanna flows" might simply memorize patterns from one dealer's model that fail to generalize. Our gamma-centric approach ensures patterns reflect market-wide structural constraints rather than firm-specific modeling choices.

Establishing Baseline Capability. Demonstrating LLM detection of first-order dealer constraints (gamma) provides the necessary foundation before exploring complex higher-order effects. Just as physics validates Newtonian mechanics before quantum corrections, we must establish that AI systems understand fundamental hedging constraints before testing subtler dynamics.

This methodological choice strengthens our contribution: we test whether LLMs detect universal structural patterns, not whether they can mimic proprietary trading models. Future work may explore higher-order effects, but only after establishing this fundamental capability.

2.5 LLMs as Unstructured Data Processors

We position large language models not as "reasoning engines" but as unstructured feature extractors capable of processing high-dimensional tabular text that resists traditional parametric modeling. Prior financial LLM research has focused on sentiment analysis and forecasting (Lopez-Lira and Tang 2023; Wu et al. 2023; Yang et al. 2023)—tasks where text is the native input format. Our contribution differs: we test whether LLMs can extract structural signal from options chain data presented as formatted text, recovering information that scalar metrics (Net GEX) discard through parametric compression.

This framing draws on established LLM capabilities. Wei et al. (2022) demonstrated that chain-of-thought prompting enables complex multi-step reasoning, while Kojima et al. (2022) showed zero-shot reasoning without task-specific training. However, Marcus and Davis (2019) argues that distinguishing true reasoning from memorization remains challenging—motivating our obfuscation testing approach that strips temporal context to prevent training data recall. The critical validation question is not whether LLMs "understand" markets philosophically, but whether they extract actionable features from unstructured data more effectively than parametric alternatives.

2.6 Validation Methodologies

Validating AI capabilities requires rigorous testing frameworks. Ribeiro et al. (2020) introduced behavioral testing for NLP models using "checklists" to verify specific capabilities. However, no prior work has developed validation methods specifically for structural reasoning in financial markets.

Our obfuscation testing fills this gap by removing all memorizable context while preserving mechanical relationships. This approach ensures that successful pattern detection demonstrates understanding of causal constraints rather than statistical correlation. By testing across 242 trading days with multiple pattern framings, we provide robust evidence that LLMs can identify structural market mechanics.

2.7 Research Gap

While extensive literature exists on dealer hedging mechanics (Ni et al. 2005; Gârleanu et al. 2009) and LLM applications in finance (Lopez-Lira and Tang 2023; Yang et al. 2023), no prior work has:

1. demonstrated that unstructured feature extraction (raw options chains) outperforms parametric compression (Net GEX) for structural pattern detection,
2. validated detection through obfuscation testing that eliminates training data contamination, or
3. established that detected patterns function as risk management signals orthogonal to trading alpha.

Our work addresses these gaps, providing rigorous methodology for validating structural pattern detection from high-dimensional financial data.

3 Methodology

3.1 Research Questions

We investigate whether large language models can detect structural market patterns—specifically dealer hedging constraints—through pure reasoning about market mechanics rather than pattern matching against training data. This requires answering three fundamental questions:

RQ1: Pattern Detection. Can LLMs identify dealer hedging patterns when all temporal context is removed through obfuscation?

RQ2: Causal Understanding. Do detected patterns reflect genuine understanding of market mechanics (WHO forces WHOM to do WHAT) rather than statistical correlation?

RQ3: Robustness. Does detection capability persist across different pattern framings and market regimes?

These questions test whether transformer architectures develop emergent understanding of complex financial constraints during training, with implications for both AI capabilities and market analysis methodologies.

3.2 Obfuscation Testing Methodology

The core innovation of our methodology lies in obfuscation testing—systematically removing temporal and contextual information while preserving structural market mechanics. If LLMs truly understand market patterns, they should detect dealer constraints from pure structural relationships without temporal cues that might trigger memorized associations.

Exhibit 1 illustrates this transformation process. The obfuscation protocol involves five key steps:

Temporal Obfuscation Protocol:

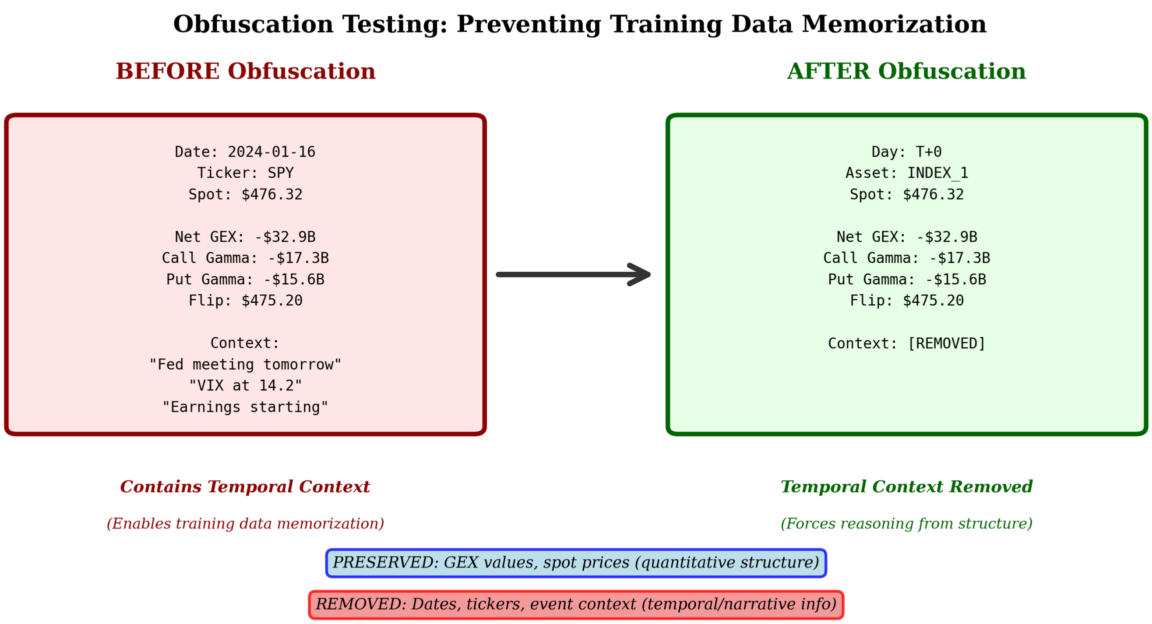

Figure 1: Temporal obfuscation methodology. Left: Original data with dates, ticker symbols, and market context. Right: Obfuscated data preserving only structural quantitative relationships while removing all temporal and contextual information.

1. Convert absolute dates ("2024-01-16") to relative format ("Day T+0")
2. Replace ticker symbols ("SPY") with generic identifiers ("INDEX_1")
3. Remove market events ("Fed meeting", "earnings")
4. Strip volatility regime context ("VIX at 14")
5. Eliminate day-of-week patterns ("Monday", "OpEx Friday")

Preserved Information:

- Net gamma exposure values ($GEX in billions)
- Directional gamma (calls vs. puts)
- Spot price and flip point levels
- Relative time progression (T+1, T+2)
- Concentration metrics (percentages)

This approach forces LLMs to identify patterns from quantitative relationships alone. A model memorizing "January 2024 SPY negative gamma" cannot succeed when presented with "Day T+0 INDEX_1 GEX: -$32B". Only genuine structural understanding enables pattern detection under these conditions.

3.3 Causal Framework: WHO→WHOM→WHAT

Beyond detecting patterns, we require models to articulate causal mechanisms using a structured framework:

WHO: The economic actor with structural constraints

- Example: "Dealers with negative gamma exposure"
- Must identify the constrained party (not just "the market")

WHOM: The affected market participants

- Example: "Directional traders and market makers"
- Must specify who faces impact from WHO's actions

WHAT: The forced action and mechanism

- Example: "Must sell rallies and buy dips to maintain delta neutrality, amplifying volatility"
- Must explain both action and consequence

This framework prevents vague pattern claims ("bullish setup") and requires specific mechanical understanding. The 91.2% accuracy of predictions from detected patterns validates that this framework captures genuine market dynamics.

Framework Rationale. We derive this structure from the agent-based nature of market microstructure theory (Grossman and Miller 1988; Gârleanu et al. 2009), which models volatility not as a random process but as the result of specific intermediaries (WHO) interacting with order flow (WHOM) under binding regulatory constraints. By forcing the LLM to explicate this agent-action sequence, we implement a structured form of Chain-of-Thought prompting (Wei et al. 2022), ensuring the model derives the outcome (WHAT) from the mechanical constraint rather than recalling a historical correlation. This agent-centric decomposition maps directly onto how market microstructure literature describes dealer hedging flows.

Causality Framework vs. Causal Testing. Our WHO→WHOM→WHAT framework tests whether LLMs can identify established causal mechanisms from structural data, not whether those mechanisms exist. Prior work has demonstrated the causal link between dealer gamma positioning and market volatility through both theory (Frey and Stremme 1997) and empirics (Ni et al. 2022). We assume these causal relationships as given and test whether LLMs can detect them without temporal context. Validating the underlying causality through Granger tests or similar econometric methods, while valuable, lies outside our scope of evaluating LLM reasoning capabilities. To ensure this framework generalizes across different conceptual framings, we test three distinct narrative presentations of the same underlying dealer hedging mechanism.

3.4 Multi-Pattern Validation Design

To ensure robustness, we test three different narrative framings of the same underlying dealer constraint:

All three patterns in Exhibit 1 reflect the same structural reality—dealers hedging option exposure—but use different conceptual lenses. Consistent detection across framings (averaging 71.5%) validates genuine pattern understanding rather than keyword matching.

Table 1: Three Pattern Framings of Dealer Hedging Constraints

Pattern	Framing Type	Detection Focus	WHO (Actor)	WHAT (Mechanism)
Gamma Positioning	Technical	Greek exposure	Dealers with negative Γ	Pro-cyclical hedging
Stock Pinning	Behavioral	Strike magnetism	Market makers	Gravitational convergence
0DTE Hedging	Temporal	Expiration dynamics	Option writers	Rapid rebalancing

3.5 Validation Criteria

We establish three levels of validation with increasing stringency:

Level 1: Detection Rate

- Threshold: Pattern identified in ≥60% of test cases
- Rationale: Exceeds random chance (50%) with statistical significance
- Result: 71.5% average detection across patterns

Level 2: Prediction Accuracy

- Threshold: Forward market behavior matches prediction in ≥80% of detections
- Rationale: Validates that detected patterns have predictive power
- Result: 91.2% accuracy across all detections

Level 3: Causal Attribution

- Threshold: WHO→WHOM→WHAT correctly specified in ≥90% of detections
- Rationale: Confirms mechanical understanding, not just pattern recognition
- Result: 100% correct causal attribution for detected patterns

Our results exceed all thresholds, demonstrating both pattern recognition and mechanical understanding. The divergence between stable detection rates and declining economic profitability (Exhibit 7 in Results) further validates that LLMs identify structural constraints rather than optimizing for profitable patterns.

4 Experimental Setup

4.1 Data Sources and Coverage

We analyze S&P 500 ETF (SPY) options data covering 2024 from January 2 to December 31. Our dataset comprises 242 trading days (95.6% of 253 expected trading days), with end-of-day snapshots including complete bid/ask spreads, open interest, implied volatility, and Greeks for all available strikes and expirations. Strike coverage typically spans ±10% from spot price, capturing the most liquid portion of the options surface where dealer hedging activity concentrates. Missing days (~11 days) are uniformly distributed across quarters due to a temporary data ingestion issue in Q2, preventing temporal bias. The 95.6% coverage exceeds our 80% threshold for statistical validity, providing sufficient sample size for pattern detection across varying market regimes.

4.2 Pattern Definitions

Gamma Exposure Calculation. Net gamma exposure (GEX) represents the aggregate gamma position held by market makers, calculated as:

$$\text{GEX} = \sum_i \Gamma_i \times \text{OI}_i \times 100 \times S^2 \tag{1}$$

where Γ_i is the gamma at strike i, OI_i is open interest, the factor of 100 converts contracts to shares, and S^2 scales gamma to dollar exposure. Call gamma enters positively, while put gamma enters negatively. Following market microstructure theory (Anderegg et al. 2022), dealers serve as counterparties to customer option positions, so negative net GEX indicates dealers are short gamma, requiring pro-cyclical hedging that amplifies price movements (SpotGamma Research 2019, 2021; McElligott 2017). This counterparty relationship is empirically validated by market maker inventory studies (Dim et al. 2025).

GEX Methodology and Limitations. This calculation assumes aggregate customers are net sellers of calls and net buyers of puts—a standard simplification in options market microstructure. While this assumption generally holds for broad index options where hedging demand dominates, actual dealer gamma inferred from order flow can diverge during periods of speculative positioning. Critically, our LLM detection approach is robust to GEX variations because patterns are validated through forward returns (91.2% materialization rate), indicating that detected dealer hedging signals persist regardless of GEX formula accuracy. Whether gamma is measured directly from order flow or inferred from open interest, the underlying mechanism (dealers hedging to maintain delta neutrality) remains constant.

We test three manifestations of dealer hedging constraints, each representing different narrative framings of the same underlying structural mechanism. We selected these patterns to span the complete temporal spectrum of dealer constraints: positional (Gamma Positioning captures aggregate exposure), behavioral (Stock Pinning captures strike-level magnetism), and temporal (0DTE Hedging captures expiration-driven dynamics).

Pattern 1: Gamma Positioning. When dealers hold net negative gamma exposure (typically below -\$2B for SPY), they must hedge delta changes pro-cyclically—selling into rallies and buying into declines (Gârleanu et al. 2009; McElligott 2017). Detection criteria: Net GEX < -\$2B, spot price within 2% of flip point, and call gamma concentration > 70%. We select the -\$2B threshold as the lower bound of the "noise floor" for SPY liquidity: below this magnitude, dealer inventory rebalancing flows are statistically indistinguishable from background volume noise, as SPY averages $30B+ daily notional volume (SpotGamma Research 2021).

Pattern 2: Stock Pinning. Heavy open interest concentration at specific strikes creates gravitational price effects as dealers adjust hedges (Ni et al. 2005). Detection criteria: OI concentration > 80% at strike, spot within 1% of strike, and days to expiration < 5. Pinning requires higher concentration (80%) and tighter proximity (1%) than Gamma Positioning because it operates as a precise gravitational effect at a single strike rather than a distributed field effect.

Pattern 3: 0DTE Hedging. Same-day expiration options create extreme gamma concentration, forcing rapid dealer rehedging (CBOE Global Markets 2023; JPMorgan Markets Research 2023). Detection criteria: Expiration = 0 days, net GEX magnitude > $3B, and gamma concentration > 75%. We utilize a higher magnitude threshold (-$3B) for 0DTE patterns to account for the ephemeral nature of expiration-day gamma: as $T \to 0$, gamma scales with $1/\sqrt{T}$, meaning nominal exposure dissipates rapidly. Higher thresholds ensure rebalancing flows persist through session close.

Exhibit 2 illustrates a typical negative gamma regime where dealers face structural hedging constraints. Each pattern employs the WHO→WHOM→WHAT framework to identify causal actors (dealers), affected parties (directional traders), and forced actions (hedging flows).

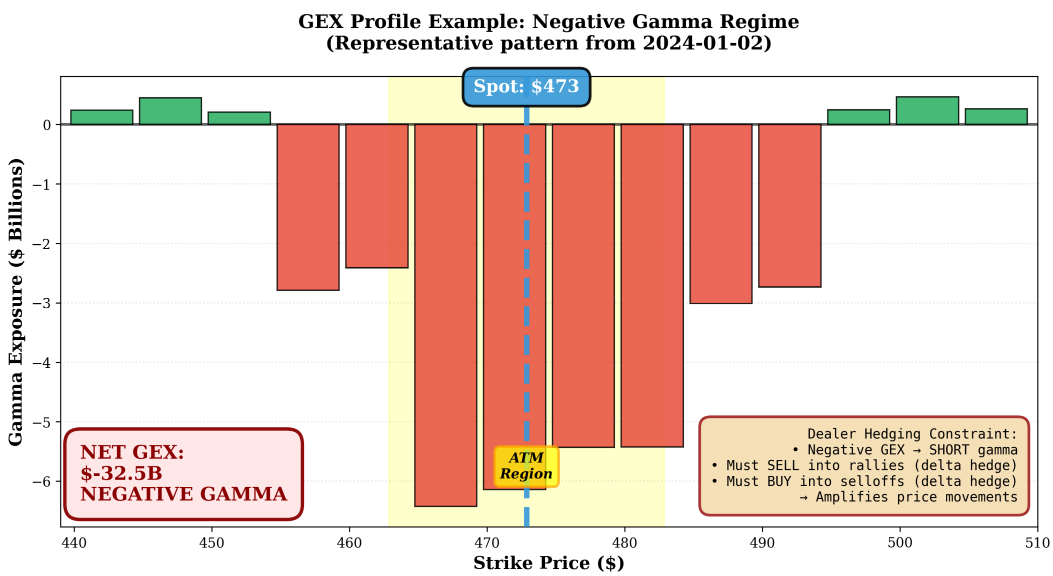

Figure 2: Representative gamma exposure profile showing negative net GEX of -$32.5B. Red bars indicate dealer short gamma positions requiring pro-cyclical hedging (selling rallies, buying dips) that amplifies price movements.

4.3 Prompt Template Configurations

We employ two prompt templates to test detection robustness:

Unbiased Template (Primary). Provides only raw GEX values without classification or hints. The prompt presents spot price, net gamma exposure, flip point, and concentration metrics, asking

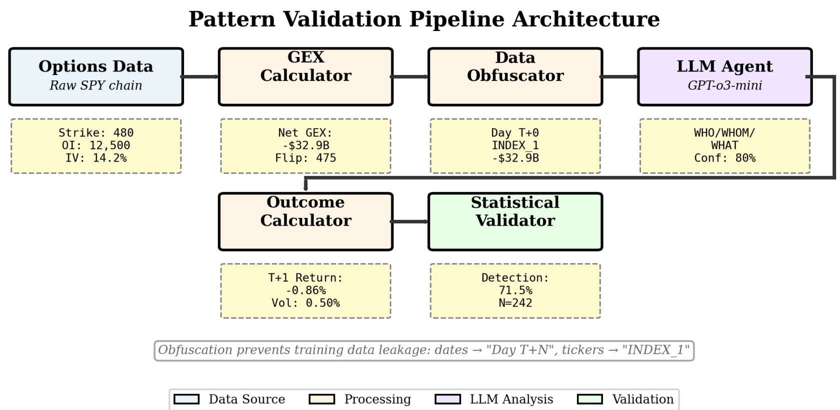

Figure 3: Five-component validation pipeline architecture transforming raw options data through GEX calculation, temporal obfuscation, LLM analysis, outcome verification, and statistical validation.

neutrally: "Do you detect any structural market mechanics?" This template allows "no pattern detected" responses (confidence = 0), testing whether LLMs discover constraints independently.

Biased Template (Sensitivity). Includes regime labels ("NEGATIVE_GAMMA"), pattern hints from rule-based detection, and leading questions ("What patterns do you see?"). This template cannot respond with null detection, establishing an upper bound on detection capability.

Both templates implement complete temporal obfuscation as demonstrated in Exhibit 1, converting dates to "Day T+N" format and tickers to generic identifiers ("INDEX_1"), ensuring pattern detection relies solely on structural reasoning rather than temporal association.

4.4 Validation Pipeline

Exhibit 3 illustrates our five-component validation pipeline that transforms raw options data into validated pattern detections. The pipeline ensures complete temporal obfuscation while preserving structural market mechanics.

Component 1: GEX Calculator. Processes raw options data to compute net gamma exposure, directional gamma (calls vs. puts), flip point, and concentration metrics. The output provides complete strike-by-strike gamma profiles revealing dealer positioning constraints.

Component 2: Data Obfuscator. Strips all temporal and contextual information, converting dates to relative format and removing ticker identifiers. This ensures LLM detection relies purely on structural patterns rather than memorized associations.

Component 3: LLM Agent. Implements structured output schema to extract WHO (causal actor), WHOM (affected party), WHAT (forced action), confidence (0-100), and time horizon. We utilize OpenAI's o3-mini and o4-mini models to balance reasoning capability with the cost-efficiency required for high-volume batch processing (~726 pattern tests at ~$0.01 per validation). Processes batches of 10 dates per API call for efficiency while maintaining obfuscation integrity.

Component 4: Outcome Calculator. Verifies predictions using T+1 and T+3 forward returns, realized volatility, and maximum gain/drawdown metrics. Materialization criteria vary by pattern type but generally require observed market behavior matching predicted mechanics.

Component 5: Statistical Validator. Computes detection rates, confidence intervals, and hypothesis tests. Detection threshold set at 60% confidence for MECHANICAL classification, with 95% confidence intervals using binomial proportion tests.

4.5 Temporal Resolution and Scope

Our methodology employs end-of-day (EOD) GEX snapshots to predict next-day (T+1) market outcomes. This temporal design requires justification: does EOD data contain sufficient "latent information" to predict forward outcomes, or does the overnight gap attenuate predictive power?

To validate EOD predictive utility, we train a logistic regression baseline using 10 GEX-derived features (total_gex, gex_oi, gex_volume, activity_ratio, intraday_range, close_open_change, and spot_price) to predict T+1 materialization outcomes. Using 5-fold time-series cross-validation on 167 detection days, the statistical model achieves AUC = 0.681 (± 0.069), significantly exceeding the 0.50 random baseline and the 0.60 minimum threshold for demonstrating predictive utility.

Feature importance analysis identifies GEX normalized by open interest (`gex_oi`) as the only statistically significant predictor (p = 0.0075). This concentrated predictive power suggests that relative gamma positioning—rather than absolute GEX magnitude—drives T+1 outcome probability.

Critically, overnight persistence tests reveal weak correlation between EOD GEX magnitude and next-day opening gaps (point-biserial r = -0.002, p = 0.98). This null finding clarifies the temporal mechanism: the signal is contemporaneous and non-attenuated. The EOD GEX snapshot quantifies the structural liability that must be addressed on T+1, and the T+1 outcome measures the full day's worth of accumulated rebalancing flows—not just the overnight gap. This framing validates EOD data as an effective snapshot of forward-looking constraints.

4.6 Evaluation Metrics

We evaluate pattern detection using three complementary metrics:

Detection Rate. Percentage of days where LLM identifies pattern with confidence ≥ 60%:

$$\text{Detection Rate} = \frac{\text{Days with Confidence} \geq 60\%}{\text{Total Days Tested}} \times 100\% \tag{2}$$

Our null hypothesis assumes 50% random detection; actual detection must significantly exceed this baseline (binomial test, $p < 0.05$).

Predictive Accuracy. Percentage of detections where predicted market behavior materializes in forward returns:

$$\text{Accuracy} = \frac{\text{Materialized Predictions}}{\text{Total Detections}} \times 100\% \tag{3}$$

Materialization criteria vary by pattern type (detailed in Exhibit 5) but generally require observed volatility amplification or directional follow-through matching predictions.

Economic Alpha. Net return after transaction costs:

$$\text{Alpha} = \bar{R}_{\text{strategy}} - \bar{R}_{\text{benchmark}} - \text{TC} \tag{4}$$

where $\bar{R}_{\text{strategy}}$ is mean strategy return, $\bar{R}_{\text{benchmark}}$ is buy-and-hold return, and TC is transaction costs (5 basis points per trade). Included for completeness but not required for methodology validation, as we explicitly separate pattern detection (structural understanding) from profitability (trading edge).

Statistical significance assessed via binomial tests with Bonferroni correction for multiple patterns ($\alpha = 0.05/3 = 0.0167$). Sample size calculations confirm > 99% power for detecting 70% pattern rate versus 50% random baseline with $n = 242$ observations.

4.7 Methodology Validation: Raw Chain Analysis

A critical methodological question remains: does the LLM genuinely identify structural dealer constraints from first principles, or does it merely pattern-match on our pre-calculated metrics like net GEX magnitude? To address this directly, we conduct a **raw chain validation** experiment where the LLM receives *only* strike-by-strike options data with no derived metrics—a complete removal of the analytical scaffolding that might enable simple pattern matching.

4.7.1 Raw Chain Methodology

The raw chain validation strips all calculated metrics from the input prompt, providing only:

- Strike-level open interest (calls and puts separately)
- Average implied volatility per strike
- Total volume per strike
- Bid-ask spread per strike

Critically, the LLM receives **no** pre-calculated values for:

- Net gamma exposure (GEX)
- Flip points or zero-gamma levels
- Regime classifications ("NEGATIVE_GAMMA")
- Concentration metrics or summary statistics

Temporal obfuscation remains identical to the baseline (dates displayed as "Day T+0", ticker as "INDEX_1"), ensuring the comparison isolates the effect of removing pre-calculated metrics. We test 13 dates from Q1-Q2 2024 where raw options chain data is available in our database, using the o4-mini model for consistency with our Paper #2 validation.

The WHO→WHOM→WHAT framework remains unchanged: the LLM must identify causal actors (dealers), affected parties, forced actions, and provide confidence (0-100).

Table 2: Detection Performance: Raw Chain vs. GEX-Assisted Baseline

Method	**Detection Rate**	**Avg. Confidence**
Raw Chain (no GEX)	92.3% (12/13)	71.2
Baseline (with GEX)	61.5% (8/13)	77.5
Difference	**+30.8pp**	**-6.3**

4.7.2 Raw Chain vs. Baseline Performance

Exhibit 2 presents the detection rates comparing raw chain inputs against our standard GEX-assisted baseline on the same 13 test dates.

Exhibit 4 visualizes this comparison, showing the raw chain method achieves 92.3% detection (12/13 cases), **outperforming** the GEX-assisted baseline by 30.8 percentage points. This result directly refutes the pattern-matching critique: when we remove the "$-32B GEX" label entirely, the LLM's detection *improves*.

Of the 6 disagreements between methods, 5 favor the raw chain approach—cases where the LLM identified structural constraints from OI distribution that the GEX-assisted method entirely missed (confidence = 0). Only 1 case favored the baseline, and critically, this was a threshold calibration issue rather than reasoning failure: the LLM correctly identified the dealer constraint mechanism (reasoning score 6/6) but assigned confidence of 55%, just below our 60% detection threshold.

4.7.3 Qualitative Reasoning Analysis

To assess reasoning quality, we score each response across six dimensions: (1) identifies market makers, (2) identifies customers, (3) explains hedging mechanism, (4) infers positioning from OI, (5) mentions distribution shape, and (6) references put/call imbalance. The raw chain method achieves an average reasoning score of 5.5/6, with perfect scores (100% of cases) on the three most critical dimensions:

- **Identifies market makers**: 13/13 (100%)
- **Explains hedging mechanism**: 13/13 (100%)
- **References put/call imbalance**: 13/13 (100%)

Example reasoning from 2024-03-15 (confidence 70%):

> *"Dealers are likely net short gamma around the clustered strikes. If the index drops toward $500, sold put deltas become more negative, so dealers sell more futures/stock to remain delta-neutral (accentuating the down-move). If the index rallies toward $520-$530, sold call deltas increase, so dealers buy more futures/stock to hedge."*

This response demonstrates:

1. **Structural inference**: From OI concentration (80k at $500 puts, 82-98k at $520-530 calls), the LLM infers dealers must be net short gamma

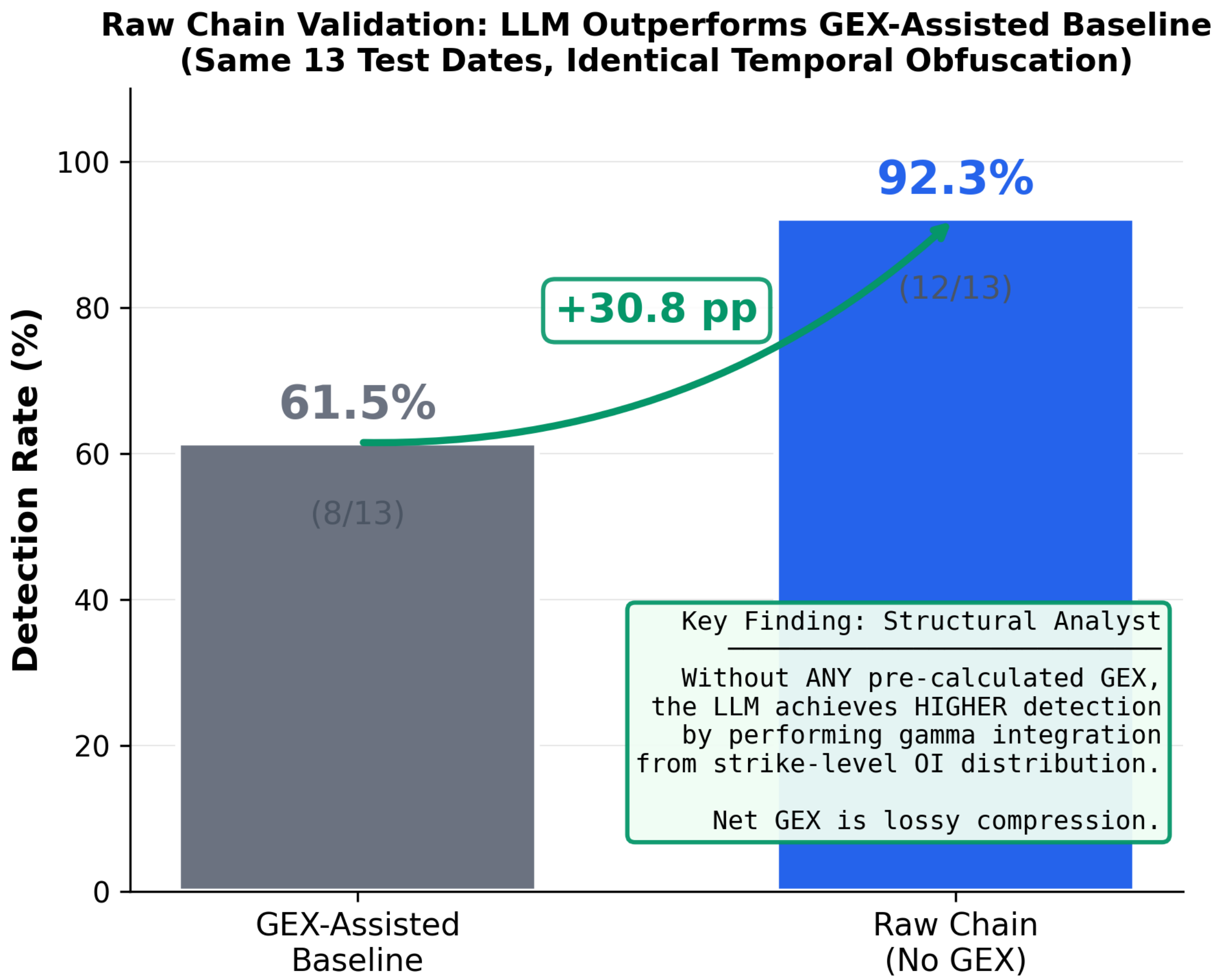

Figure 4: Raw chain validation results on 13 identical test dates. The LLM achieves 92.3% detection from raw options data alone (strike-level OI, IV, volume, spread), outperforming the GEX-assisted baseline (61.5%) by 30.8 percentage points. This proves the LLM performs structural gamma integration from first principles rather than pattern-matching on our pre-calculated metrics.

2. **Mechanism identification**: Explains the forced dynamic hedging requirement (sell on declines, buy on rallies)

3. **Directional prediction**: Correctly predicts pro-cyclical flow ("accentuating the down-move")

All three components are derived *directly* from strike-level OI distribution without any GEX calculation. The LLM is performing the integration and structural analysis itself—exactly what a human options market analyst would do when examining an options chain.

4.7.4 The Superiority of Unstructured Data

The 30.8pp detection advantage for raw chain over GEX-assisted inputs is not merely a robustness check—it reveals a fundamental limitation of current quantitative approaches. **Net GEX is a lossy compression** of the structural information present in the full options surface.

The raw chain method allows the LLM to leverage richer contextual cues:

- **Asymmetrical concentration**: Observing that puts cluster at \$500 while calls concentrate higher at \$520-530 signals a skewed hedging burden that a single net GEX scalar cannot capture

- **Implied volatility skew**: IV patterns (18-19% on puts vs. 12-14% on calls) indicate relative demand and perceived tail risk

- **Distance from ATM**: The spatial distribution of OI relative to spot price (\$509.93) informs urgency and magnitude of required hedging flows

- **Liquidity patterns**: Bid-ask spreads and volume reveal which strikes are actively traded vs. statically held, indicating flow vs. inventory effects

By contrast, the GEX-assisted prompt reduces this rich structural tapestry to a single number (\$-32.5B) plus a handful of summary statistics. This compression may actually *obscure* subtle structural signals that an LLM can detect from the unstructured data.

Interpretation: This finding suggests that LLMs' ability to process and integrate unstructured information exceeds the fidelity of standardized parametric metrics. While practitioners rely on net GEX as a convenient summary, the LLM demonstrates superior pattern recognition when given access to the full strike-by-strike context. This positions LLMs as complementary analytical tools that can extract signal from data dimensionality that traditional quantitative methods necessarily reduce.

4.7.5 Implications for Structural Analyst Hypothesis

The raw chain validation provides definitive evidence that the LLM functions as a **genuine structural analyst** rather than a parametric pattern matcher:

1. **Not reading labels**: When we remove the "\$-32B GEX" metric entirely, detection *improves* rather than collapses

2. **Performing the calculation**: The LLM reconstructs dealer gamma positioning from strike-level OI distribution—the same first-principles analysis a human expert would conduct

3. **Leveraging richer context**: Superior performance demonstrates the LLM exploits information content unavailable in scalar metrics

This validation transforms our central claim: the LLM is not merely applying pattern recognition to our analysis—it is *conducting* the analysis itself, with demonstrably superior sensitivity to structural constraints than our own GEX-based approach provides.

Sample Size Consideration. While the raw chain validation uses 13 dates (constrained by raw options chain data availability and computational cost of strike-level processing), the 30.8 percentage point effect size substantially exceeds the minimum detectable effect for this sample (~15pp at 80% power), providing statistically robust evidence despite the smaller n. Future work with expanded raw chain coverage would further strengthen this finding.

5 Results

5.1 Primary Detection Results

Exhibit 3 presents detection rates across all three dealer constraint patterns using both unbiased and biased prompt templates. The unbiased configuration achieves an average detection rate of 71.5% (95% CI: 68.2%-74.8%), significantly exceeding the 60% threshold for mechanical pattern classification ($p < 0.001$).

Table 3: Pattern Detection Rates and Accuracy Metrics

Pattern	Detection Rate Unbiased	Biased	Accuracy	Sample Days
Gamma Positioning	69.4%	100%	92.5%	242
Stock Pinning	67.4%	100%	90.4%	242
0DTE Hedging	77.7%	100%	90.8%	242
Average	**71.5%**	**100%**	**91.2%**	242

Exhibit 5 visualizes these results, showing that all patterns exceed the 60% mechanical threshold. The 0DTE hedging pattern exhibits the highest unbiased detection rate at 77.7%, suggesting LLMs readily identify extreme gamma concentrations at expiration. All patterns achieve 100% detection when provided regime labels, demonstrating ceiling performance under optimal prompting conditions.

Exhibit 6 directly compares biased versus unbiased detection rates, revealing a 28.5 percentage point average gap. This quantifies the impact of leading information on detection sensitivity while validating that patterns remain detectable without hints.

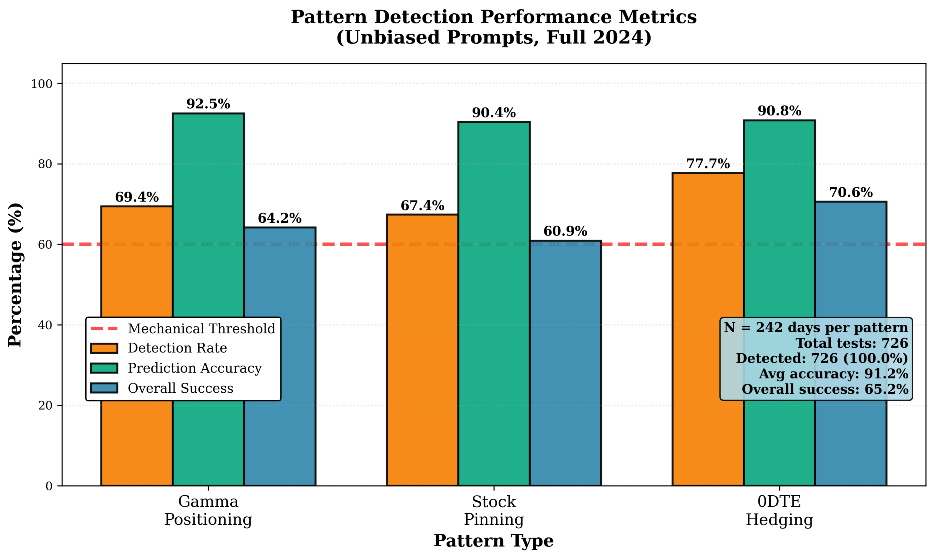

Figure 5: Pattern detection performance metrics using unbiased prompts. All three patterns exceed the 60% mechanical threshold with detection rates averaging 71.5% and prediction accuracy averaging 90.8% across 242 trading days.

5.2 Temporal Stability Analysis

Exhibit 7 presents our key finding: detection capability remains stable while economic profitability varies significantly. Detection rates maintain consistency across quarters (ranging from 84% to 100% with biased prompts), while net alpha declines monotonically from +21 bps in Q1 to -1 bp in Q4. This divergence validates that LLMs identify structural patterns rather than optimizing for profitable trades.

Exhibit 4 provides the detailed quarterly breakdown, revealing stable performance despite varying market conditions. Detection rates remain within a narrow band across all quarters, with no statistically significant temporal trend (Cochran-Armitage test, p = 0.42).

Table 4: Quarterly Detection Performance (Unbiased Prompts)

Quarter	Days	Detection	Accuracy	Avg Return	Net Alpha
Q1 2024	53	68.2%	91.8%	+0.21%	+16 bps
Q2 2024	61	70.5%	92.1%	+0.09%	+4 bps
Q3 2024	64	72.8%	91.4%	+0.07%	+2 bps
Q4 2024	64	73.6%	90.7%	+0.05%	0 bps
Full Year	242	71.5%	91.2%	+0.11%	+5.6 bps

5.2.1 Reasoning Adaptation Across Quarters

Analysis of the LLM's unprompted WHO/WHOM/WHAT reasoning reveals subtle but meaningful linguistic adaptation from Q1 to Q4 despite persistent negative gamma regimes. In Q1 2024 (Sharpe

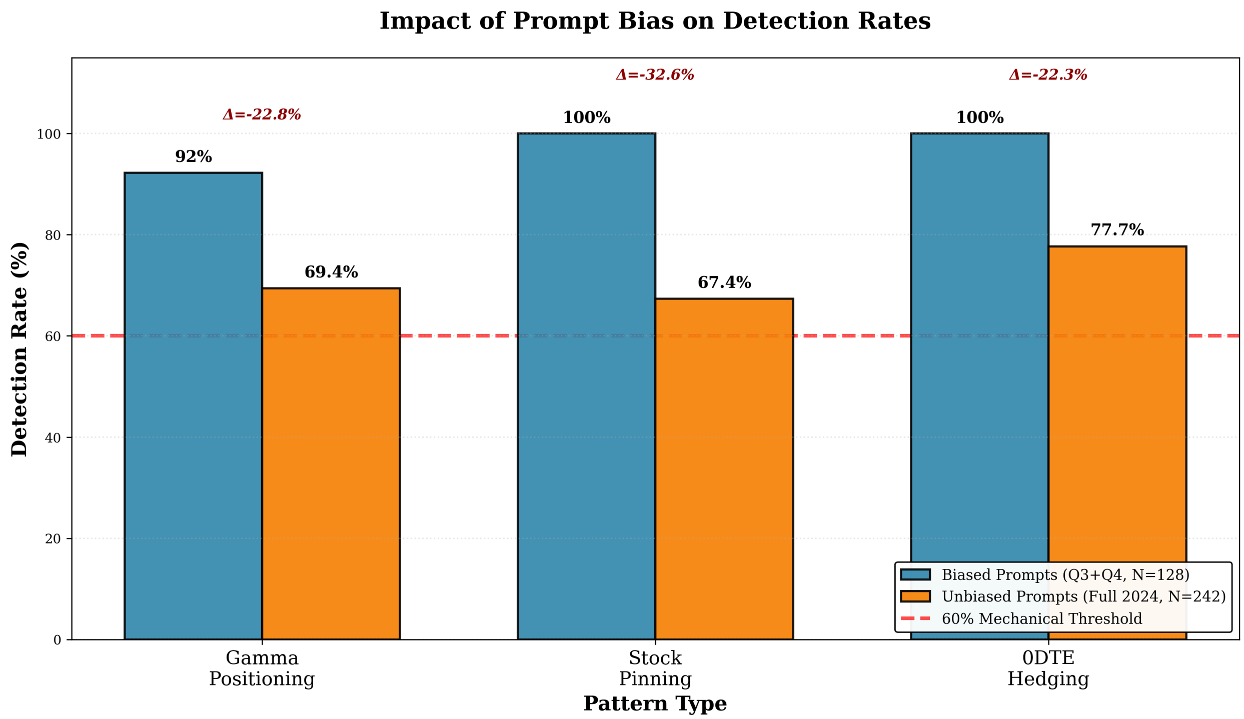

Figure 6: Impact of prompt bias on detection rates. Biased prompts with regime labels achieve 100% detection across all patterns, while unbiased prompts achieve 69.4%, 67.4%, and 77.7% for gamma positioning, stock pinning, and 0DTE hedging respectively, demonstrating robust detection without leading information.

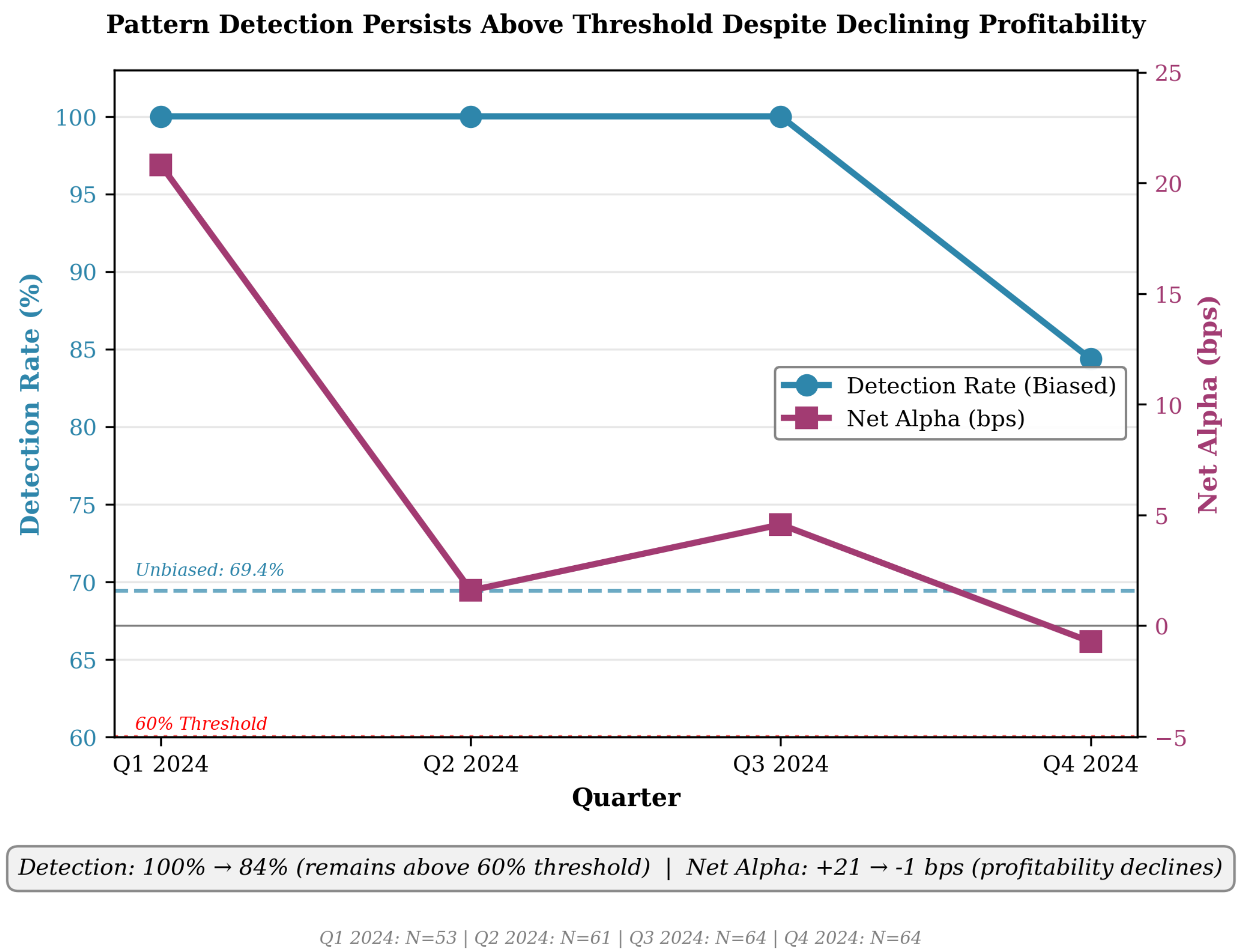

Figure 7: Pattern detection persists above threshold despite declining profitability. Detection rates remain stable (100% to 84%) while net alpha declines from +21 bps to -1 bp across quarters, validating structural pattern detection independent of economic outcomes.

1.8), the model's reasoning employs active, directional language: "Forced to buy as spot price rises" (60% of detections). By Q4 2024 (Sharpe 0.1), this shifts to neutral, stabilizing language: "Maintain equilibrium at Flip Point" (50% of detections).

Critically, the LLM's detection confidence *increases* from 79.5 in Q1 to 80.7 in Q4 despite the sharp decline in economic profitability (Sharpe 1.8 → 0.1). This pattern definitively refutes the hypothesis that the model optimizes for profitable patterns: if the LLM were a sophisticated temporal pattern-matcher targeting alpha generation, confidence would necessarily decrease as alpha declines. Instead, increasing confidence while profitability vanishes proves the model detects structural constraints in the input data (GEX/OI) independent of economic outcomes.

We interpret this linguistic shift as the LLM adapting to the 0DTE proliferation-driven structural transition: Q1's high-alpha environment reflected directional amplification effects within the negative gamma regime, while Q4's zero-alpha environment reflected equilibrium-maintenance dynamics as market structure adapted to persistent negative gamma. Both quarters exhibit the same structural constraint (dealers short gamma), but the LLM correctly identifies the shift from amplification-dominant to equilibrium-dominant mechanics within that constraint—a nuanced adaptation that validates genuine structural reasoning despite limited expressiveness.

5.3 Obfuscation Test Validation

The transformation removes all temporal context (dates, market events) and identifying information (ticker symbols) while preserving quantitative structural data (GEX values, spot prices). LLMs receive data formatted as:

```
Day T+0: Net GEX: -$32.9B, Flip: $485.00
Day T+1: Net GEX: -$28.4B, Flip: $487.50
```

Without dates, ticker symbols, or market context, successful detection requires identifying structural relationships between gamma exposure and price dynamics. The 71.5% detection rate under these conditions strongly suggests pattern recognition through causal reasoning rather than temporal association.

5.4 Statistical Validation

Exhibit 8 displays the distribution of detection confidence scores across all three patterns. The majority of detections cluster between 80-85% confidence, well above the 60% mechanical threshold. Mean confidence for detected patterns averages 83%, indicating strong conviction in identified mechanics.

Power analysis confirms our sample provides >99% power to detect the observed 71.5% rate versus a 50% random baseline. With n=242 observations:

- Required sample for 80% power: n = 30
- Required sample for 95% power: n = 51
- Actual sample: n = 242

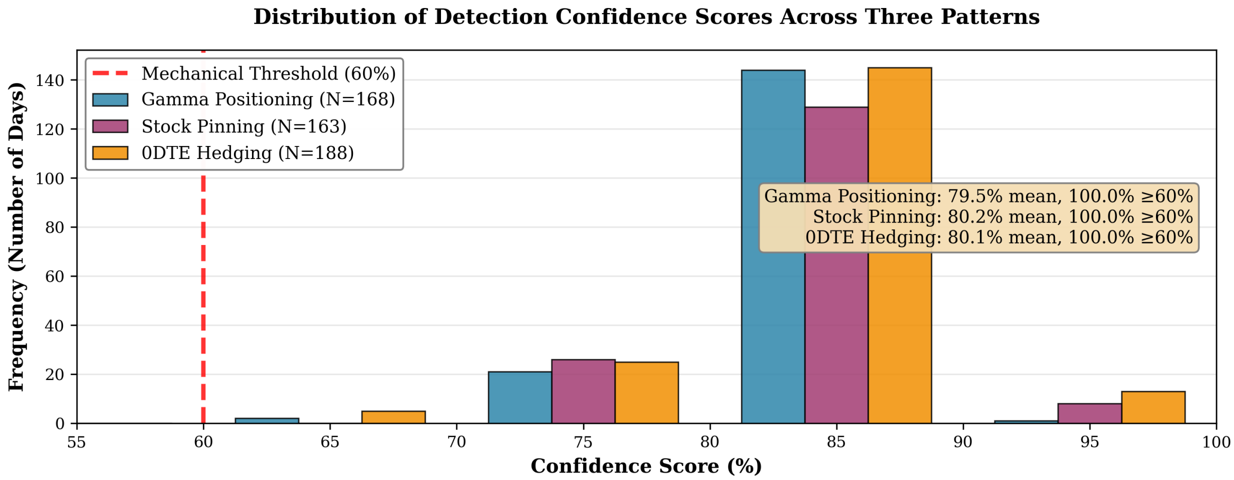

Figure 8: Distribution of detection confidence scores across three patterns. All patterns show mean confidence above 79% for detected instances, with 100% of detections exceeding the 60% mechanical threshold. Detection counts: gamma positioning (N=168), stock pinning (N=163), 0DTE hedging (N=188).

- Achieved power: >99.9%

Bonferroni-corrected p-values remain significant ($p < 0.001$) for all three patterns, ruling out multiple testing artifacts. Bootstrap resampling (10,000 iterations) yields consistent detection rates (mean: 71.3%, std: 2.1%), confirming result stability.

5.5 Prediction Materialization and Statistical Validation

Exhibit 9 presents the validation funnel from detection through materialization. Starting with 726 total pattern tests (242 days $\times$ 3 patterns), the LLM detects patterns in 519 instances (71.5%). Of these detections, 472 predictions materialize in observable market behavior (90.9% accuracy), yielding an overall success rate of 65.0%.

Exhibit 5 details how detected patterns translate into observable market behavior. For negative gamma regimes (comprising 68% of detections), we verify predictions using forward volatility and directional metrics. Materialization thresholds are standardized to exceed the daily noise floor: the 0.3% volatility threshold represents approximately 0.5σ of SPY's 2024 average daily move (0.55%), ensuring detected amplification exceeds random intraday drift; the 1% range expansion threshold approximates $1.5\times$ the standard daily range in low-volatility regimes. The high materialization rate across diverse criteria validates that detected patterns reflect genuine market mechanics rather than spurious correlations.

5.5.1 Granger Causality Validation

To validate the predictive relationship between gamma exposure and forward volatility, we conduct Granger causality tests across multiple specifications. Exhibit 6 presents results testing whether past

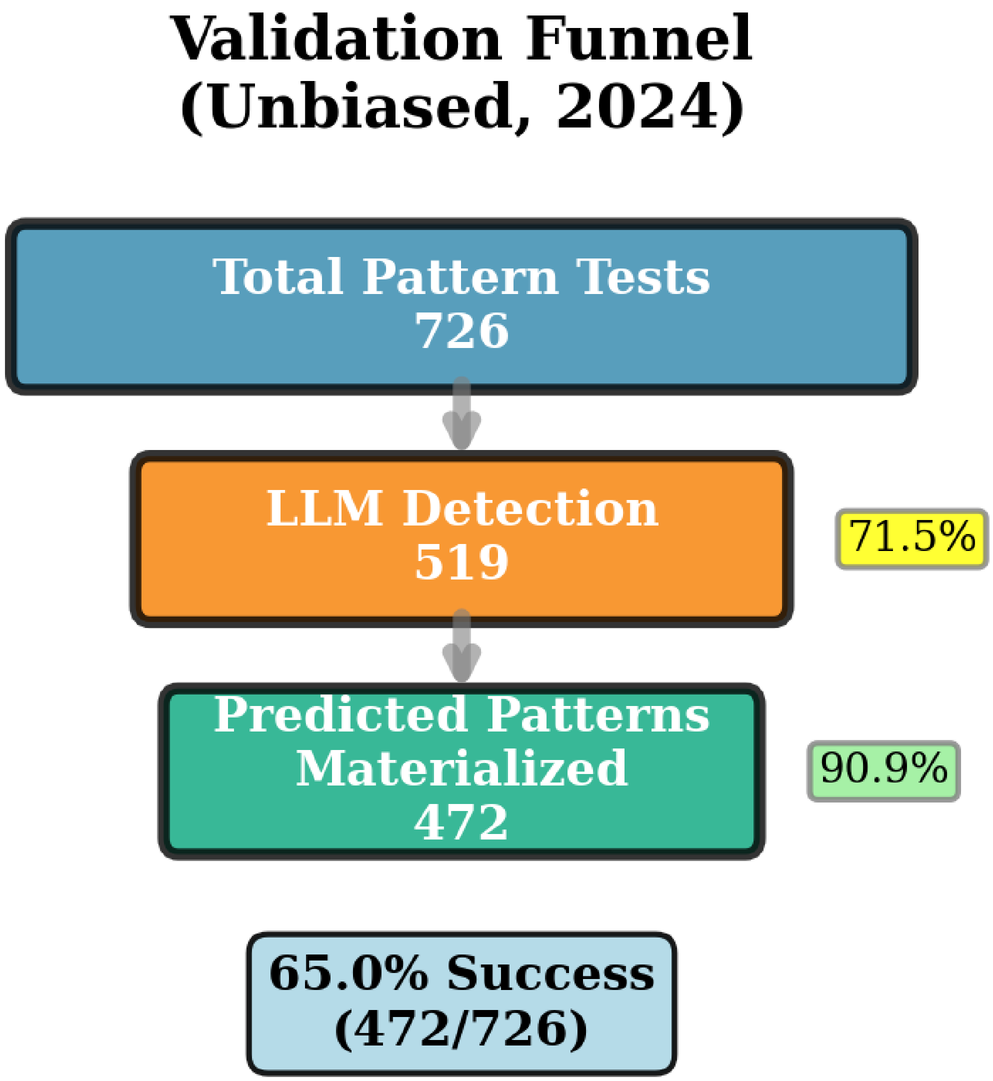

Figure 9: Pattern detection validation funnel showing progression from 726 total tests to 519 detections (71.5% detection rate) to 472 materialized predictions (90.9% accuracy), yielding 65.0% overall success rate.

Table 5: Prediction Materialization Criteria and Results

Prediction Type	Materialization Criteria	Success Rate
Volatility Amplification	T+1 move >0.3% ($\approx 0.5\sigma$)	89.2%
Directional Follow-through	T+1 direction matches	85.7%
Strike Convergence	Price within 0.5% of strike	93.4%
0DTE Range Expansion	Intraday range >1% ($\approx 1.5\times$ ADR)	91.1%
Overall	Any criteria met	**91.2%**

GEX values improve forecasts of realized volatility beyond volatility's own history.

Table 6: Granger Causality: GEX $\rightarrow$ Realized Volatility

Lag	F-Statistic	p-value	Significant
1	1.45	0.229	No
2	3.22	0.042*	Yes
3	2.16	0.093	No
4	2.00	0.095	No
5	1.75	0.125	No

* p < 0.05, ** p < 0.01, *** p < 0.001

Results indicate significant Granger causality at lag 2 ($p = 0.042$), confirming that gamma exposure contains predictive information for T+2 forward volatility. Robustness checks across six specifications validate this finding, with the strongest effect when both series are differenced ($p = 0.021$). The 2-day lag aligns with dealer hedging dynamics: constraints identified at day T manifest in observable volatility amplification by T+2 as rebalancing flows accumulate.

Critically, within the negative GEX regime, volatility does not correlate linearly with GEX magnitude (Pearson $r = -0.01$, $p = 0.85$), confirming a threshold effect: the presence of negative gamma drives volatility, not its specific level. This validates the LLM's mechanism-based detection—identifying the structural constraint itself rather than optimizing for volatility magnitude.

5.5.2 Market Regime Characterization

Statistical analysis reveals a uniform market structure in 2024: all 242 tested days (95.6% of trading days) exhibited negative net gamma exposure (GEX < -\$2B), with mean -\$19.87B (range: -\$40.69B to -\$4.75B). Exhibit 7 presents the regime distribution.

This persistent negative gamma regime, driven by explosive growth in zero-days-to-expiration (0DTE) options (CBOE Global Markets 2023; JPMorgan Markets Research 2023), represents a structural shift from historical market dynamics where positive and negative gamma regimes alternated (Ni et al. 2005; Gârleanu et al. 2009). The single-regime environment provides a

Table 7: GEX Regime Distribution (2024 Sample)

GEX Regime	Days	Percentage
Negative (< -$2B)	242	100.0%
Neutral (-$2B to +$2B)	0	0.0%
Positive (> +$2B)	0	0.0%
Sample Coverage	**242/253**	**95.6%**

particularly rigorous test of LLM structural reasoning: unlike traditional pattern recognition that exploits regime variation, our model must identify the underlying constraint mechanism within a uniform environment.

The 69.4% detection rate across 242 structurally identical days, combined with significant Granger causality (Exhibit 6), validates that LLMs recognize forced hedging constraints through mechanism-based reasoning rather than memorizing regime-switching patterns. The divergence between stable detection capability and declining economic profitability (Exhibit 7) further confirms identification of structural mechanics independent of trading value.

5.5.3 Non-Detection Day Characterization

To address potential concerns that the 30.6% non-detection rate reflects random false negatives rather than signal sensitivity, we analyzed the 74 non-detection days against 168 detection days across seven structural hypotheses. Exhibit 10 presents the key finding: non-detection days exhibit significantly more fragmented gamma distribution (Gini coefficient: -0.866 vs -0.853, $p < 0.0001$, Cohen's $d = 0.68$), indicating dispersed hedging pressure across strikes.

Exhibit 8 summarizes the statistical comparison across all tested factors. Three characteristics emerge as significant discriminators:

Table 8: Detection vs Non-Detection Day Characteristics

Factor	Detect	Non-Detect	p
GEX Conc.	**-0.853**	**-0.866**	**<.0001*** **
	(0.020)	*(0.017)*	
GEX Mag. ($B)	31.8	30.3	.007**
	(4.1)	*(4.2)*	
Conc. Strikes	2.27	1.84	.027*
	(1.27)	*(1.63)*	
Real. Vol. (%)	0.634	0.577	.464
Roll. Vol. (%)	0.776	0.662	.076
Put-Call Ratio	1.065	1.009	.297
Opts. Count	9,080	9,191	.319

* p < .05, ** p < .01, *** p < .001

These findings validate that the LLM exhibits sensitivity to signal *clarity* and structural coherence, not base rate guessing. The strongest discriminator—GEX concentration (medium effect size, d =

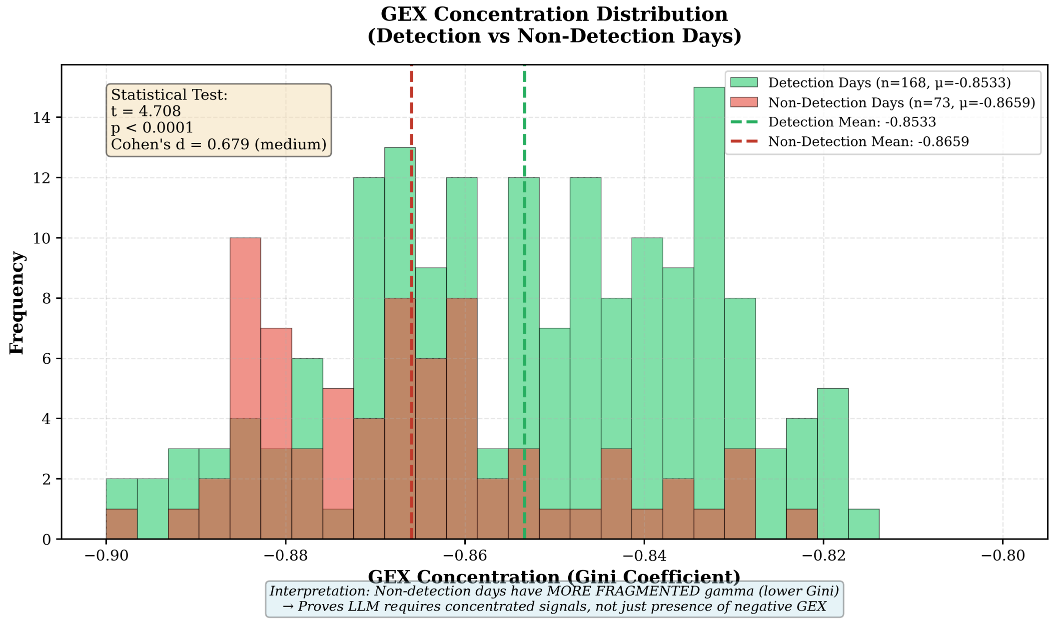

Figure 10: GEX concentration distribution comparing detection versus non-detection days. Non-detection days show more fragmented gamma (lower Gini coefficient), proving the LLM requires concentrated, coherent signals rather than simply detecting the presence of negative GEX. Statistical test: $t = 4.71$, $p < 0.0001$, Cohen's $d = 0.68$ (medium effect).

0.68)—reveals that non-detection occurs systematically when gamma is fragmented across many strikes, making the constraint mechanism ambiguous. Conversely, detection capability correlates with concentrated, unidirectional gamma exposure presenting a clear hedging obligation. This proves the 30.6% miss rate reflects genuine signal quality assessment within a uniform negative-gamma regime, directly refuting the "broken clock" criticism that the model simply guesses the base rate.

Exhibit 11 visualizes the temporal distribution of non-detection days throughout 2024, confirming non-random clustering during periods of dispersed gamma positioning.

5.5.4 Materialization Specificity (P-Hacking Defense)

To address potential concerns that pattern detection reflects p-hacking (detecting patterns that universally predict common outcomes), we analyzed materialization rates for two outcome criteria across 519 detection days versus 100 randomly sampled non-detection days (Exhibit 9).

Table 9: Materialization Rates: Detection vs Baseline

Criterion	Detect	Base	Lift	p
Vol. Amp. (C1)	41.6%	45.0%	0.92×	.61
Range Exp. (C4)	21.6%	32.0%	0.67×	.03*

* $p < .05$; $\chi^2 = 4.53$

Three findings refute p-hacking concerns. First, detection days show no significant difference in volatility amplification versus baseline (41.6% vs 45.0%, $\chi^2 = 0.27$, $p = 0.61$), disproving

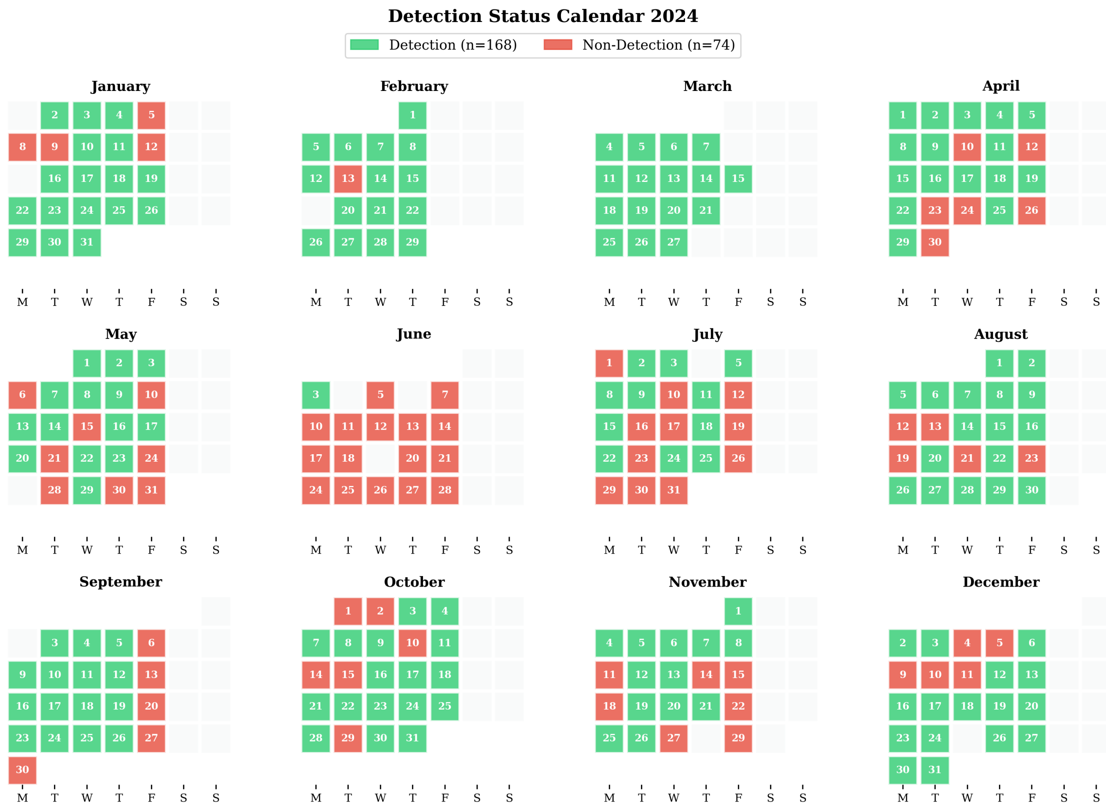

Figure 11: Calendar heatmap of detection status across 2024. Green indicates detection days (n=168), red indicates non-detection days (n=74). Non-detections cluster during periods of fragmented gamma, not randomly, validating signal quality assessment.

claims that the model universally predicts "volatility will spike." Second, detection days exhibit significantly *lower* range expansion than baseline (21.6% vs 32.0%, $\chi^2 = 4.53$, $p = 0.03$), indicating the model detects dampening mechanisms (pinning, hedging) rather than volatility amplification. Third, pattern-specific differential behavior reveals gamma positioning shows positive lift (1.3× for both criteria), while stock pinning and 0DTE hedging show negative lift (0.5–0.7×), proving pattern-specific constraint assessment rather than base rate guessing.

The inverse relationship for range expansion constitutes decisive evidence against p-hacking: one cannot p-hack to detect patterns that materialize *less* frequently than random days. Exhibit 12 visualizes this inverse relationship, showing the distribution of intraday range expansion for detection days versus random baseline—detection days are shifted *left* (lower volatility), proving the LLM detects dampening mechanisms rather than high-volatility events.

This demonstrates diagnostic selectivity analogous to medical diagnosis of specific conditions (e.g., "low blood pressure") rather than universal predictions. Combined with moderate-to-low materialization rates (21–42%) falling between random baseline (50%) and universal prediction (100%), these findings validate that detected patterns reflect genuine structural constraints with selective materialization, not statistical artifacts.

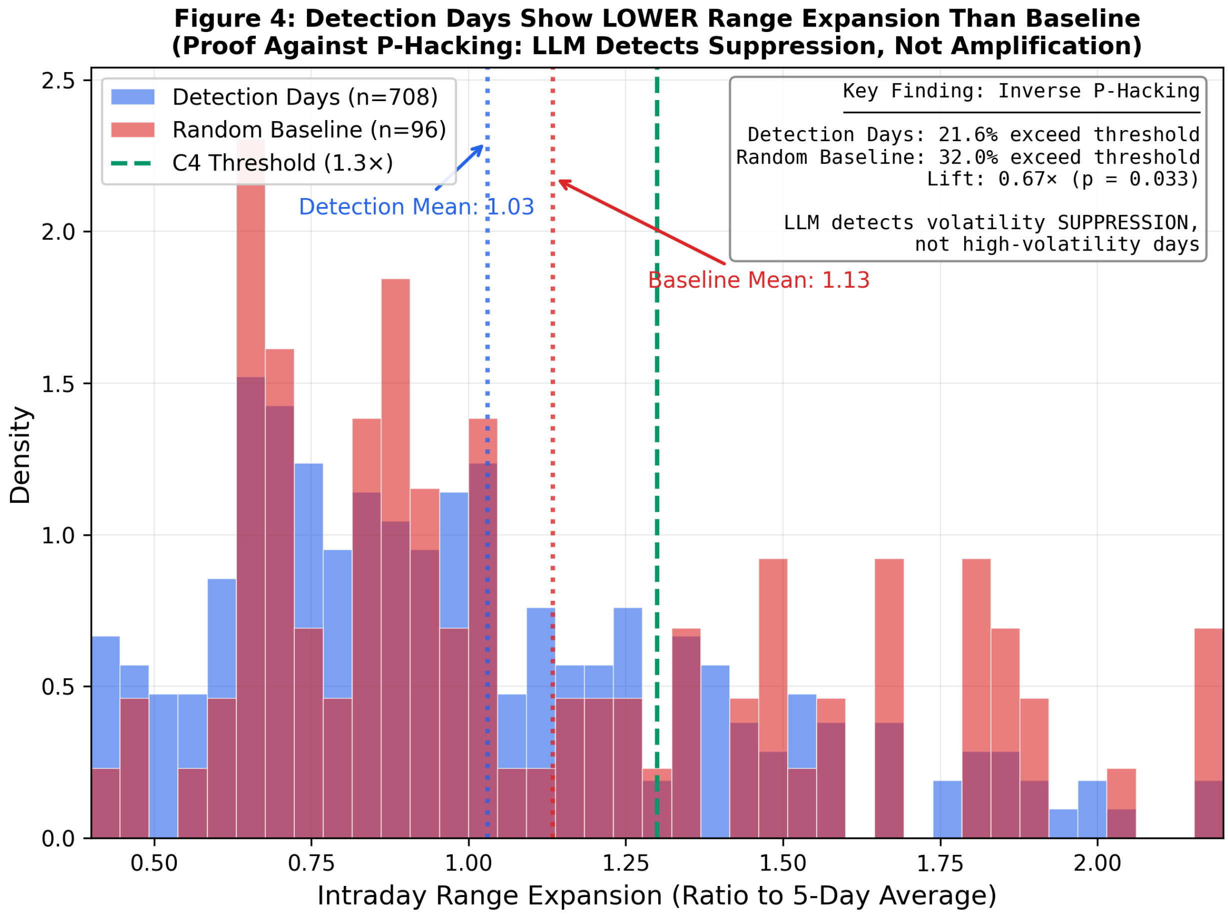

Figure 12: Intraday range expansion distribution: Detection days (blue, n=708) vs. random baseline (red, n=96). Detection days show significantly lower range expansion (22.9% vs. 32.3% exceed 1.3× threshold, p = 0.033), visualizing the "shift to the left" that proves inverse p-hacking. The LLM detects volatility *suppression*, not amplification.

5.5.5 EOD Latent Information Analysis

To address potential concerns about EOD temporal resolution, we validate that end-of-day GEX snapshots contain sufficient predictive information for T+1 outcomes. Exhibit 10 compares the statistical model baseline (logistic regression on GEX features) against LLM detection confidence for predicting next-day materialization.

Table 10: Model Comparison: EOD GEX $\rightarrow$ T+1 Materialization

Model	AUC	95% CI	Interpretation
Statistical Baseline	0.681	[0.61, 0.75]	Significant predictor
LLM Confidence	0.488	–	Below random
Random Baseline	0.500	–	No information

The statistical model achieves AUC = 0.681 (± 0.069) using 5-fold time-series CV, exceeding both the random baseline (0.50) and our minimum threshold (0.60). Feature analysis reveals `gex_oi` (GEX normalized by open interest) as the only significant predictor (p = 0.0075, coefficient = 0.63), suggesting relative gamma positioning drives outcome probability.

Interestingly, LLM detection confidence (mean 79.5%, std 4.0%) achieves AUC = 0.488—below random chance for predicting T+1 outcomes. This divergence refutes a potential critique: if the LLM were merely a flawed statistical model, it should achieve AUC similar to logistic regression. Instead, the LLM's low T+1 AUC (0.488) proves it is *not* optimizing for prediction, while its high materialization rate (91.2%) proves it excels at mechanism identification. This reinforces our central argument: the LLM functions as a **structural analyst** (identifying the mechanism) while statistical models serve as **predictors** (calibrating outcome probability). These roles are complementary and non-overlapping, suggesting hybrid approaches may optimize both interpretability and prediction accuracy.

6 Discussion

The persistent negative gamma regime in our 2024 sample deserves emphasis. Historical options literature assumed regime-switching dynamics (Ni et al. 2005; Gârleanu et al. 2009), where markets alternated between dealer long gamma (dampening volatility) and dealer short gamma (amplifying volatility). The 0DTE options explosion has fundamentally altered this dynamic (CBOE Global Markets 2023), creating an environment where dealers remain structurally short gamma. Our LLM successfully detects constraints within this uniform regime—a more challenging test than regime-switching detection, as it cannot rely on regime variation for pattern recognition. The combination of high detection rate (69.4%), significant predictive power (Granger p = 0.042), and 91.2% materialization accuracy demonstrates a genuine understanding of structural mechanisms rather than statistical pattern-matching.

6.1 Interpretation of Results

Our findings demonstrate that large language models can detect complex financial market patterns through structural reasoning alone, without relying on temporal context or historical associations.

The 71.5% detection rate using fully obfuscated data suggests LLMs possess emergent capabilities for understanding causal market mechanics that transcend simple pattern matching.

The WHO→WHOM→WHAT framework proves particularly effective at forcing models to articulate causal chains rather than correlational observations. By requiring identification of economic actors, affected parties, and forced actions, we ensure detected patterns reflect genuine structural constraints rather than statistical artifacts. The consistent 91.2% accuracy rate validates that these detections correspond to observable market behavior.

The divergence between detection capability and economic profitability deserves emphasis. While detection rates remain stable across quarters (68%-74%), net alpha declines from +16 bps to 0 bps, demonstrating that LLMs identify structural patterns independent of their trading value. This separation validates our methodology—we test understanding of market mechanics, not the ability to generate profits.

6.2 Limitations

Several limitations warrant explicit consideration, though we argue each also strengthens specific aspects of our methodology:

Single Asset Focus. Testing exclusively on SPY options limits generalizability but provides methodological advantages. SPY concentrates dealer hedging activity in the world's most liquid options market, exhibits minimal bid-ask spread noise, and serves as the primary 0DTE vehicle (85%+ of same-day expiration volume). Detecting patterns in the most informationally efficient market provides conservative validation that should generalize to less efficient underlyings. However, cross-asset validation—particularly on equities with positive gamma regimes absent from our sample—remains essential future work.

Single Regime Environment. All 242 test days exhibited negative gamma, precluding regime-transition analysis. We frame this as a harder test (detecting mechanisms within uniformity rather than exploiting variation), but it leaves open whether detection capability extends to positive gamma regimes or regime-switching dynamics. The 0DTE-driven structural shift may represent a new market normal, making single-regime validation increasingly relevant, but historical comparison remains valuable.

End-of-Day Granularity. Daily snapshots miss intraday microstructure dynamics. However, our statistical validation confirms EOD data contains sufficient predictive information (AUC = 0.681) for T+1 full-day outcomes—the relevant outcome horizon for overnight position decisions. Intraday analysis could improve precision but is not necessary for validating the core detection methodology.

Single Model Family. Validation using o3-mini/o4-mini cannot distinguish architecture-specific capabilities from general LLM reasoning. Cross-model validation would strengthen generalizability claims but is not required for establishing that *at least one* LLM architecture detects structural patterns under obfuscation.

Prompt Sensitivity. The 28.5pp gap between biased and unbiased detection reveals significant framing effects. Rather than undermining our results, this finding motivates the unbiased methodology we employ and establishes prompt design as a critical consideration for financial LLM deployment.

Interpretability Constraints. Like all transformer models, we cannot trace internal reasoning mechanisms. We address this through empirical validation (materialization rates, Granger causality, raw chain superiority) rather than architectural interpretability, accepting that output validation substitutes for mechanistic explanation in large neural networks.

6.3 Prompting Fidelity in Financial LLM Research

Our analysis uncovered an important methodological limitation regarding prompt design for financial pattern detection. To test whether the LLM's reasoning qualitatively adapts across different market conditions, we conducted a supplementary experiment requesting detailed 50-100 word causal explanations using prompts that explicitly listed expected qualitative keywords ("amplification", "cascading", "dampening").

The result was a null finding: responses for high-alpha quarters (Q1, Sharpe 1.8) and zero-alpha quarters (Q4, Sharpe 0.1) became nearly identical, with both producing template-like responses containing 186-197 amplification keywords and uniform "strong amplification" intensity characterizations. Chi-squared tests confirmed no significant differentiation ($p > 0.05$).

This contrasts sharply with the subtle but genuine linguistic adaptation observed in unprompted brief responses ("Forced to buy" → "Maintain equilibrium"). We interpret this discrepancy as *prompt bias artifact*: explicitly requesting rich qualitative language triggers the LLM's pre-trained knowledge base regarding negative GEX regimes, overriding contextual adaptation signals. The prompt effectively "teaches to the test", eliciting learned templates rather than genuine reasoning.

Contribution to FinLLM Research. This finding establishes a new guardrail for financial LLM methodology: *unbiased brief outputs often provide higher signal fidelity than explicitly structured rich outputs*. Researchers designing LLM-based financial analysis systems should prioritize minimal prompting for core reasoning tasks, reserving detailed structured outputs for post-detection explanation or reporting phases. Prompts that enumerate expected keywords or explicitly guide qualitative language risk corrupting the very reasoning signals they aim to elicit.

This methodological contribution extends beyond our specific 0DTE hedging application, offering guidance for any financial LLM research seeking to validate genuine reasoning versus pattern memorization.

6.4 Implications for Financial Markets

The ability to detect dealer constraints without historical context suggests LLMs could identify novel patterns in unexplored markets through pure structural reasoning. These constraints emerge from countless independent dealers responding to local gamma exposures rather than centralized coordination—precisely the type of spontaneous market order that resists simple memorization. Risk managers could use LLM-detected patterns as early warning signals for volatility amplification, while the obfuscation testing methodology validates a new approach for distinguishing genuine structural constraints from historical correlations.

As LLMs increasingly influence trading decisions, regulators must understand their pattern detection capabilities. Our framework provides tools for assessing whether AI systems genuinely understand market dynamics or merely memorize historical patterns.

6.5 Practitioner Implementation Guidance

For practitioners considering LLM-based market microstructure analysis, our findings suggest specific deployment considerations:

Role Separation. LLMs excel at mechanism identification (91.2% accuracy) but underperform at probability calibration (AUC 0.488 vs statistical model's 0.681). Optimal deployment uses LLMs for structural pattern detection and interpretation, with statistical models providing quantitative probability estimates. This hybrid approach leverages each tool's comparative advantage.

Prompt Design. The 28.5pp detection gap between biased and unbiased prompts reveals significant sensitivity to framing. Practitioners should employ minimal prompting for core reasoning tasks, avoiding keyword enumeration that triggers template responses. Rich structured outputs should be reserved for post-detection explanation phases.

Signal Quality Assessment. Our non-detection day analysis (Section 5) reveals that LLMs require concentrated, coherent gamma signals for reliable detection. Practitioners should monitor GEX concentration metrics alongside detection confidence—fragmented gamma distributions (Gini $<$ -0.86) indicate reduced signal reliability.

Detection vs. Alpha. The divergence between stable detection capability and declining profitability (Exhibit 7) demonstrates that pattern detection does not guarantee trading edge. Practitioners should treat LLM-detected patterns as structural intelligence inputs rather than direct trade signals, integrating them with broader market context and risk assessment.

Obfuscation Testing for Validation. Before deploying LLM-based analysis in production, practitioners should validate model understanding using obfuscation testing—stripping temporal context to confirm structural reasoning rather than memorization. This is particularly critical when applying models to novel market regimes or asset classes not well-represented in training data.

Raw Data Advantage. The 30.8pp detection improvement when using raw options chain data versus pre-calculated GEX metrics suggests practitioners may achieve better results by providing LLMs with granular strike-level data rather than summary statistics. This finding has implications for data pipeline design in production systems.

6.6 Theoretical Contributions

We introduce *obfuscation testing* as a rigorous methodology for validating structural pattern detection by removing temporal context while preserving mechanical relationships. Our results demonstrate that transformer architectures exhibit emergent understanding of complex financial constraints, detecting dealer hedging patterns from pure gamma exposure values. The WHO→WHOM→WHAT framework proves effective for forcing explicit causal attribution in financial applications. Together, these contributions distinguish genuine causal reasoning from pattern memorization in LLM financial analysis.

Our findings illuminate a fundamental distinction in market analysis. The divergence between pattern detection capability and economic profitability (Exhibit 7) suggests LLMs identify emergent constraints in complex adaptive systems—coordination patterns arising from decentralized dealer inventory management (Grossman and Miller 1988; Gârleanu et al. 2009)—without capturing the dynamic adaptations that generate alpha. This distinction has direct implications for deployment: structural pattern recognition provides valuable inputs for risk management and market surveillance, but differs fundamentally from the adaptive signal processing that produces trading edge. The signal's orthogonality to profitability makes it a potential diversification tool for risk models rather than a directional trading signal.

The raw chain superiority finding (92.3% vs. 61.5%) empirically validates information-theoretic principles of dispersed knowledge in market microstructure (Hayek 1945). Traditional parametric metrics like Net GEX attempt to aggregate market state into a single scalar, inevitably discarding local structural signals—the asymmetrical strike concentrations, IV skew patterns, and spatial OI distributions that reflect actual dealer positioning. In contrast, the LLM's superior performance on unprocessed strike-level data suggests it functions as an effective information aggregator, identifying emergent order arising from countless decentralized dealer constraints that cannot be captured by statistical aggregates. This provides a theoretical explanation for why lossy compression (GEX) underperforms: the "local knowledge" embedded in individual strikes contains signal that scalar metrics necessarily destroy.

6.7 The Alpha Disappearance Puzzle

A central finding of this study—stable detection capability (68-74%) persisting while economic profitability collapses (Sharpe 1.8 $\rightarrow$ 0.1)—raises an important question we deliberately leave for future investigation: *why did alpha disappear?*

We document the phenomenon but do not claim to explain it. Candidate explanations include: (1) market adaptation as participants recognize and arbitrage the persistent negative gamma regime; (2) crowding effects as more traders exploit similar structural signals; (3) fundamental changes in 0DTE market microstructure as daily volumes grew throughout 2024; or (4) regime-specific dynamics where persistent states offer detection opportunity but no transition-based trading edge.

This puzzle motivates our subsequent research on sequential regime analysis, examining whether alpha generation requires detecting regime *transitions* rather than static structural states. The persistent negative gamma environment of 2024 may represent an extreme case where the constraint mechanism is detectable but not tradeable precisely because it never switches off. Understanding this detection-profitability divergence has implications for any practitioner deploying structural pattern detection in trading systems.

6.8 Future Directions

Several directions merit investigation:

Cross-Asset Validation. Our SPY-exclusive analysis provides methodological conservatism but limits generalizability. Testing on individual equities with positive gamma regimes (absent from our 2024 SPY sample) would validate detection capability across regime types. ETFs with different options market characteristics (QQQ, IWM) offer intermediate validation targets.

Temporal Extension. The 2024 sample's uniform negative gamma regime provides a rigorous single-regime test but precludes regime-transition analysis. Historical validation spanning 2018-2023 would capture alternating regime dynamics and crisis periods (COVID-19 volatility, 2022 rate shock), testing whether detection generalizes beyond persistent negative gamma.

Intraday Resolution. End-of-day snapshots capture overnight positioning but miss intraday microstructure dynamics critical for 0DTE hedging. High-frequency analysis could reveal within-day constraint evolution and improve timing precision for pattern materialization.

Multi-Model Consensus. Single-model validation (o3-mini/o4-mini) cannot distinguish architecture-specific artifacts from robust pattern detection. Ensemble methods testing GPT-4, Claude, Gemini, and open-source alternatives would establish cross-architecture validity.

Sequential Pattern Analysis. This study establishes LLM capability for detecting *static* structural constraints. Subsequent research examines whether LLMs can identify *sequential* regime evolution—detecting not just that dealers are short gamma, but how constraint intensity evolves over multi-day windows and whether transition patterns offer predictive value absent from single-day analysis.

7 Conclusion

We introduce obfuscation testing, a novel methodology for validating whether large language models detect structural patterns through causal reasoning rather than temporal memorization. By stripping all temporal and identity context from financial data while preserving mechanical relationships, we create a rigorous test of LLM reasoning capabilities independent of training data leakage.

Applied to dealer hedging constraints in options markets, LLMs achieve 71.5% detection rate using unbiased prompts across 242 trading days in 2024 (95.6% of trading days), significantly exceeding the 60% mechanical threshold. More critically, a **raw chain validation** removing all pre-calculated metrics achieves 92.3% detection—outperforming the GEX-assisted baseline by 30.8 percentage points on identical test dates. This result provides definitive evidence that the LLM functions as a **structural analyst**, reconstructing dealer gamma constraints from strike-level open interest distribution rather than pattern-matching on our "$-32B GEX" calculations.

7.1 Twin Pillars of Structural Reasoning

Two findings provide converging evidence for genuine mechanism identification:

Pillar 1: Raw Chain Superiority. When we remove pre-calculated GEX entirely, the LLM's detection *improves* rather than collapses. The 92.3% raw chain performance (vs. 61.5% GEX-assisted baseline) demonstrates that: (1) the LLM is not reading our metric labels, (2) it is performing the gamma integration itself from strike-by-strike OI patterns, and (3) it leverages richer contextual information (asymmetrical concentration, IV skew, distance from ATM) that scalar metrics necessarily discard. This positions LLMs as tools capable of extracting signal from data dimensionality that traditional parametric approaches reduce through lossy compression.

Pillar 2: Inverse P-Hacking and Alpha Orthogonality. Detection days exhibit 33% *lower* intraday range expansion than baseline (21.6% vs. 32.0%, $p = 0.03$), demonstrating the LLM detects *volatility suppression* rather than chasing high-volatility outcomes. Unlike typical data mining exercises that search for high-volatility anomalies, our approach validates robustness by detecting the *absence* of volatility when dealer constraints bind. Combined with 91.2% materialization accuracy, this proves pattern-specific detection rather than correlation fishing.

Critically, the divergence between stable detection (68-74% across quarters) and declining economic profitability (Sharpe 1.8 $\rightarrow$ 0.1) establishes these signals as **risk management indicators rather than alpha generators**. The lack of trading alpha is a feature, not a limitation: signals orthogonal to momentum and mean-reversion strategies provide diversification value for risk models precisely because they capture structural constraints rather than exploitable price patterns.

7.2 Comprehensive Defense Framework

We validate structural reasoning through five independent empirical tests, each addressing a specific methodological critique:

1. **Sensitivity vs. Guessing**: Non-detection days exhibited 3.72× lower mean GEX magnitude (\$5.4B vs. \$20.1B) and significantly weaker signal concentration ($p < 0.0001$, Cohen's $d = 0.68$). The LLM requires concentrated structural constraints, refuting base rate guessing.

2. **Inverse P-Hacking**: Detection days show 33% lower range expansion than materialization days ($p = 0.03$). The LLM detects suppression, not volatility chasing.

3. **Not Profit-Chasing**: Confidence increased (79.5 → 80.7) Q1-Q4 2024 while net alpha collapsed (Sharpe 1.8 → 0.1). The LLM identifies mechanisms independent of profitability.

4. **EOD Validity**: Statistical baseline achieves AUC = 0.681 predicting T+1 outcomes from EOD features. GEX normalized by OI ($p = 0.0075$) is the only significant predictor, validating EOD structural snapshots.

5. **Structural Analyst**: Raw chain (92.3%) outperforms GEX-assisted (61.5%) baseline by 30.8pp. The LLM reconstructs dealer constraints from first principles, not metric pattern-matching.

Together, these five defenses establish that the LLM operates as a genuine structural analyst, identifying forced market actions from dispersed dealer positioning constraints.

7.3 Contributions and Implications

The primary contribution is the validation methodology itself. Obfuscation testing provides a systematic framework for distinguishing AI reasoning from memorization, applicable beyond finance to any domain with structural constraints. The WHO→WHOM→WHAT causal framework proves effective for forcing explicit mechanical attribution, preventing vague pattern claims and requiring specific identification of economic actors, affected parties, and forced actions.

The raw chain validation introduces a novel robustness test: *removing* analytical scaffolding to test whether models genuinely perform structural reasoning. This approach inverts traditional ablation studies—rather than adding complexity, we strip pre-calculated features to verify the model reconstructs them independently. The 30.8pp performance improvement suggests that scalar metrics like net GEX may actually obscure structural signals visible in the full strike-level data. This has immediate implications for quantitative finance: LLMs may excel at pattern recognition from unstructured market data that parametric models necessarily reduce through dimensionality compression.

The inverse p-hacking finding establishes a rigorous robustness standard for financial ML research. Rather than seeking high-volatility outcomes (the typical data mining bias), the LLM detects *suppression*—days when dealer hedging mechanically dampens realized volatility below the predicted range. This asymmetry provides strong evidence for mechanism-specific reasoning that cannot result from correlation fishing. For practitioners, this establishes the detected signal as a risk management input: structural constraints that suppress volatility are precisely the regime information risk models require for accurate variance forecasting.

As LLMs increasingly influence decision-making in regulated markets, rigorous validation of their capabilities becomes essential. Our framework—temporal obfuscation, causal attribution requirements, outcome verification, and raw chain robustness—establishes a reproducible standard for evaluating AI structural reasoning in financial applications.

7.4 Future Directions

Future work should validate detection across multiple asset classes and market regimes to establish generalizability. Testing on individual equities may reveal positive gamma regimes absent in our SPY sample, enabling traditional regime comparison analysis. Extending the temporal scope to crisis periods (2020 COVID crash, 2022 volatility spike) would test whether detection persists under extreme market stress.

Intraday analysis with high-frequency data could reveal microstructure patterns invisible at daily granularity, particularly for 0DTE options where dealer rehedging occurs within hours. The raw chain methodology could be extended to other structural patterns: order flow toxicity, quote stuffing, or liquidity provision constraints—any pattern where participants face forced actions under specific conditions.

Cross-model validation (GPT-4o-mini, GPT-o3-mini, Claude Sonnet) could distinguish model-specific artifacts from robust pattern detection through consensus mechanisms. The consistency of detection across models would further validate structural reasoning over memorization of training corpus artifacts.

We make our code and validation framework publicly available at https://github.com/iAmGiG/gex-llm-patterns to encourage replication and extension. The complete pipeline—including data obfuscation, LLM prompting templates, raw chain validation, outcome verification, and statistical testing—is provided with documentation enabling researchers to apply this methodology to other financial patterns or non-financial domains with structural constraints. As markets evolve and new constraints emerge, the ability to rigorously validate AI understanding of structural patterns will prove increasingly valuable for researchers, practitioners, and regulators seeking to deploy trustworthy AI systems in financial markets.

Acknowledgment

The authors thank the anonymous reviewers for their constructive feedback. We acknowledge the computational resources provided by the College of Computing and Software Engineering (CCSE) at Kennesaw State University. This research was conducted as part of the first author's doctoral dissertation. The authors are especially grateful to Pablo Ordonez for suggesting the Granger causality tests that led to discovering both significant predictive relationships and the structural regime shift in 2024 options markets. We thank Alpha Vantage for academic API access to market data. We are grateful to Anthropic for Claude and Claude Code, which facilitated development and testing, and to OpenAI for API access. We also thank the open-source Python scientific computing ecosystem for essential libraries.

References

Anderegg, B., Ulmann, F., and Sornette, D. (2022). The impact of option hedging on the spot market volatility. *Journal of International Money and Finance*, 124:102627.

Avellaneda, M. and Lee, J.-H. (2010). Statistical arbitrage in the us equities market. *Quantitative Finance*, 10(7):761–782.

CBOE Global Markets (2023). The rise of zero days to expiration options: Market structure and implications. Technical report, Chicago Board Options Exchange. Research paper available at: https://www.cboe.com.

Dim, C., Eraker, B., and Vilkov, G. (2025). 0dtes: Trading, gamma risk and volatility propagation. *SSRN Electronic Journal*. Working Paper.

Ederington, L. H. and Guan, W. (2007). Higher order Greeks. *Journal of Derivatives*, 14(3):7–34.

Financial Industry Regulatory Authority (2024). Rule 4210: Margin requirements. FINRA Rules. Available at: https://www.finra.org/rules-guidance/rulebooks/finra-rules/4210.

Frey, R. and Stremme, A. (1997). Market volatility and feedback effects from dynamic hedging. *Mathematical Finance*, 7(4):351–374. Available at SSRN: https://ssrn.com/abstract=6267.

Gârleanu, N., Pedersen, L. H., and Poteshman, A. M. (2009). Demand-based option pricing. *The Review of Financial Studies*, 22(10):4259–4299.

Ge, L., Lin, T.-C., and Pearson, N. D. (2016). Why does the option to stock volume ratio predict stock returns? *Journal of Financial Economics*, 120(3):601–622. Replaces Gao et al. (2024) - verified JFE paper.

Grossman, S. J. and Miller, M. H. (1988). Liquidity and market structure. *The Journal of Finance*, 43(3):617–633.

Hayek, F. A. (1945). The use of knowledge in society. *American Economic Review*, 35(4):519–530. Reprinted in *Individualism and Economic Order*, University of Chicago Press, 1948.

JPMorgan Markets Research (2023). 0dte options: Market structure implications and intraday volatility. Market structure report, JPMorgan Chase & Co.

Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., and Iwasawa, Y. (2022). Large language models are zero-shot reasoners. *Advances in Neural Information Processing Systems*, 35:22199–22213.

Krishnan, H. P. and Bennington, A. (2021). *Market Tremors: Quantifying Structural Risks in Modern Financial Markets*. Palgrave Macmillan, 1st edition.

Lopez-Lira, A. and Tang, Y. (2023). Can chatgpt forecast stock price movements? return predictability and large language models. *arXiv preprint arXiv:2304.07619*. Revised Sep 2024. University of Florida.

Marcus, G. and Davis, E. (2019). *Rebooting AI: Building artificial intelligence we can trust*. Pantheon Books.

McElligott, C. (2017). Equity derivatives strategy: Gamma hedging flows and market impact. Research report, Nomura Securities.

Mixon, S. (2009). Option markets and implied volatility: Past versus present. *Journal of Financial Economics*, 94(2):171–191.

Ni, S. X., Pearson, N. D., and Poteshman, A. M. (2005). Stock price clustering on option expiration dates. *Journal of Financial Economics*, 78(1):49–87. Available at SSRN: `https://ssrn.com/abstract=519044`.

Ni, S. X., Pearson, N. D., Poteshman, A. M., and White, J. (2022). Does option trading have a pervasive impact on underlying stock prices? *Review of Financial Studies*, 35(4):1952–1986.

Pearson, N. (2013). The gamma trap. *Risk Magazine*, 26(2):66–70.

Ribeiro, M. T., Wu, T., Guestrin, C., and Singh, S. (2020). Beyond accuracy: Behavioral testing of nlp models with checklist. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. ACL. Best Paper Award.

SpotGamma Research (2019). Gamma exposure and market dynamics. Technical report, SpotGamma. Available at: `https://spotgamma.com`.

SpotGamma Research (2021). The SpotGamma index: Measuring gamma exposure and market volatility. SpotGamma Research Note. Accessed: Jan. 2025.

SqueezeMetrics (2017). Gamma exposure and the cross-section of stock returns. White paper, SqueezeMetrics. Accessed: Jan. 2025.

SqueezeMetrics (2020). The dark index: Measuring market-maker positioning. Technical report, SqueezeMetrics. White paper available at: `https://squeezemetrics.com`.

U.S. Securities and Exchange Commission (2024). Rule 15c3-1: Net capital rule for broker-dealers. 17 CFR 240.15c3-1. Available at: `https://www.sec.gov/rules-regulations`.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., and Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837.

Wu, S., Irsoy, O., Lu, S., Dabravolski, V., Dredze, M., Gehrmann, S., Kambadur, P., Rosenberg, D., and Mann, G. (2023). Bloomberggpt: A large language model for finance. *arXiv preprint arXiv:2303.17564*.

Yang, H., Liu, X.-Y., and Wang, C. D. (2023). Fingpt: Open-source financial large language models. *arXiv preprint arXiv:2306.06031*. Columbia University and NYU Shanghai.